CHAPTER 10

The Generator Coordinate Method

10.1 Introduction

The microscopic description of collective motion has up to now been based on a generalized product state chosen so as to include as many correlations as possible. Using this picture as a basis, further correlations were then taken into account by including boson-like excitations. In principle, it is possible to obtain the exact eigenstates in the boson picture, but in practice we are restricted to a limited number of bosons. As a result, we are able to describe harmonic vibrations in nuclei that are stiff against deformations and anharmonic effects in the transition region. However, for nuclei that are so soft that they are just about to collapse (i.e., in the vicinity of a phase transition), boson expansion techniques up to a definite order may no longer be sufficient. We have to do at least a partial summation up to infinite order. It is clear, for example, that the nuclear fission process, where really *large amplitude collective motions* (LACM) are involved, is also a candidate for a theory that has to go beyond that considered so far. To handle these situations, essentially two theories have been proposed: the *generator coordinate method* (GCM) and the *time dependent Hartree–Fock theory* (TDHF). In this chapter, we will consider the GCM. It is a fully quantum mechanical method and consists of a construction of a linear superposition of many different product wave functions and is, in a sense, a generalization of the LCAO (linear combination of atomic orbital) method in molecular physics for the case of continuously labeled basis functions [MEL 76]. The method can be used for a variety of problems,

such as the proper treatment of symmetries in many-body wave functions (see Chap. 11), the cluster representation of nuclei [WM 66, BW 68], and the scattering of composite particles on nuclei [Fl 72, TRG 73, FHW 74, BBB 75]. Although in the back of our minds we shall restrict the application of the GCM to LACM, many of the problems we will discuss are equally valid for any other bound state problem. We will, however, omit all questions specifically concerning the treatment of unbound states (for a review, see [Wo 75]).

In Section 10.2 we will introduce the general concept and apply it in Section 10.3 to the Lipkin model. The close relationship between GCM and boson expansions is treated in Section 10.4. A very instructive example is the harmonic oscillator (Sec. 10.5). Complex generator coordinates are handled in Section 10.6. One of the reasons to apply the GCM to the theory of collective motion is the fact that we can derive in a fully quantum mechanical way a collective Hamiltonian which provides a microscopic connection to the collective model described in Chapter 1 (Sec. 10.7). In Section 10.8, a method is presented which allows determination of the optimal generating functions. Some applications of the GCM are discussed in Section 10.9.

10.2 The General Concept

10.2.1 The GCM Ansatz for the Wave Function

The generator coordinate method consists of a very general ansatz for a trial function. It is a continuous superposition of the so-called generating functions $|\Phi(a)\rangle$, which are labeled by an unlimited number of real or complex parameters $\{a\}=a_1,a_2,\ldots,a_i$, the so-called generator coordinates; they depend, as does the wave function in the many-body Hilbert space, on the particle coordinates

$$|\Psi\rangle=\int da\, f(a)|\Phi(a)\rangle. \tag{10.1}$$

The integral is a multidimensional one going over all real and imaginary parts of the parameters a. The name "*generator coordinate*" is somewhat misleading, because the as are not coordinates in the strict sense. The GCM does not introduce redundant coordinates. They are only a kind of continuous summation index in the expansion (10.1). The function $f(a)$ is a weight and is therefore called a *weight function*, although as we shall see, in the mathematical sense it is no wave function.

The set of generating functions $|\Phi(a)\rangle$—as well as the limits of integration in Eq. (10.1)—can be chosen to be anything we like. Depending on the nature of the functions $|\Phi(a)\rangle$ and the number of parameters a_i, the ansatz (10.1) may eventually contain the exact solution of the Schrödinger equation. In many cases we use product wave functions $|\Phi(a)\rangle$. An example is

Nilsson wave functions in a potential of deformation β (see Chap. 2). In this case, β would be the generator coordinate.

Another example is the most general product HFB-type wave functions* $|\Phi(z^*)\rangle$, not orthogonal to some basic function $|\Phi_0\rangle$. They are given by the theorem of Thouless (Sec. E.3)

$$|\Phi(z^*)\rangle = N_0 \exp\left\{ \sum_{k<k'} z^*_{kk'} \alpha_k^+ \alpha_{k'}^+ \right\} |\Phi_0\rangle, \qquad \langle\Phi|\Phi\rangle = 1 \tag{10.2}$$

where $z_{kk'}$ $(k<k')$ are the complex numbers

$$z_{kk'} = x_{kk'} + iy_{kk'}, \qquad \int dz = \int_{-\infty}^{+\infty} \prod_{k<k'} dx_{kk'}\, dy_{kk'}. \tag{10.3}$$

In Section 10.4 we will see that the exact eigenfunctions of the many-body Hamiltonian H can be described with generating functions of the type (10.2). However, the number of generator coordinates $z_{kk'}$ is extremely large in this case. For practical applications, we must reduce their number as much as possible, and it requires physical intuition to choose a set of generating functions $|\Phi(a)\rangle$ which is general enough to describe the physical situation and still simple enough to be treated by the following methods. In general, therefore, with a fixed set of generating functions we are only able to describe a certain class of eigenstates of the Hamiltonian H by an ansatz of the form (10.1).

10.2.2 The Determination of the Weight Function $f(a)$

The weight function $f(a)$ of the ansatz (10.1) is assumed to be a well behaved function of the variables a (in general square integrable), such that we can perform the following mathematical operations.† For a determination of the function $f(a)$ we use the variational principle:

$$\delta \frac{\langle\Psi|H|\Psi\rangle}{\langle\Psi|\Psi\rangle} = 0. \tag{10.4}$$

Variation with respect to $f(a)$ gives the integral equation

$$\int da' \langle\Phi(a)|H|\Phi(a')\rangle f(a') = E \int da' \langle\Phi(a)|\Phi(a')\rangle f(a'). \tag{10.5}$$

This equation has become known as the *Hill–Wheeler equation* [HW 53]. It may be formally written

$$\mathcal{H} f = E \mathcal{N} f \tag{10.6}$$

* For convenience, we use here z^* instead of z [see Eq. (10.79)].

† The reader who is interested in deeper mathematical foundations is referred to the references [La 74, La 76, TPG 77, TP 78], where the mathematical properties of the GCM techniques are treated with complete rigor.

with the overlap functions

$$\mathcal{H}(a,a')=\langle\Phi(a)|H|\Phi(a')\rangle, \qquad (10.7)$$
$$\mathcal{N}(a,a')=\langle\Phi(a)|\Phi(a')\rangle$$

as integral kernels.

Formally, it looks very similar to the diagonalization of the Hamiltonian in the nonorthogonal "basis" of the generator states $|\Phi(a)\rangle$ [Wo 70]. This is certainly true if the parameter a (we will restrict ourselves in the following to only one real parameter) has only a finite number of values $a_1,\ldots,a_M$. In this case there is a finite number of functions $|\Phi_i\rangle=|\Phi(a_i)\rangle$ that are, in general, linearly dependent and not complete. In fact, the Hill–Wheeler equation (10.5) is often solved numerically by discretization. We then immediately get a matrix equation of the form (10.6). In the general case of a continuous parameter a, however, we have to be a little careful, as we shall see in the following.

The formal solution of Eq. (10.6) is easy. We could try to invert the operator $\mathcal{N}$ and diagonalize $\mathcal{N}^{-1}\mathcal{H}$. This yields a non-Hermitian eigenvalue problem, and this is only possible if $\mathcal{N}$ has no zero eigenvalues. However, this is very often the case, since the set of $|\Phi(a)\rangle$ is, in general, linearly dependent. Therefore, we solve the problem by an orthogonalization of the set $|\Phi(a)\rangle$. There are several methods by which to proceed. The best known is the procedure of Gram–Schmidt, which corresponds to a decomposition of the matrix $\mathcal{N}$ into two triangular matrices (see Sec. 8.4.4). Another method which is also applicable to the continuous case is the so-called symmetric orthogonalization. It corresponds to a diagonalization of the Hermitian operator $\mathcal{N}$

$$\int da'\,\mathcal{N}(a,a')u_k(a')=n_k u_k(a). \qquad (10.8)$$

In the following, we assume that the spectrum n_k $(k=1,2,\ldots)$ is discrete.* Since $\mathcal{N}$ is a norm, its eigenvalues are never negative $(n_k\geqslant 0)$.

The functions $u_k(a)$ form a complete orthonormalized set in the space of the weight functions $f(a)$

$$\sum_k u_k(a)u_k^*(a')=\delta(a-a'); \qquad \int da\,u_k^*(a)u_{k'}(a)=\delta_{kk'}. \qquad (10.9)$$

The operator $\mathcal{N}$ in this space can be decomposed $(\mathcal{N}=\mathcal{N}^{1/2}\cdot\mathcal{N}^{1/2})$ with

$$\mathcal{N}^{1/2}(a,a')=\sum_k u_k(a)\sqrt{n_k}\,u_k^*(a'), \qquad (10.10)$$

*This is the case if $\mathcal{N}$ is of the Hilbert–Schmidt type (i.e., Hermitian and with a finite trace of $\mathcal{N}^+\mathcal{N}$ [RS 72]). In many practical cases the kernels $\mathcal{N}$ and $\mathcal{H}$ do not have this property [for instance, Eq. (10.49)]. However, it has been shown that by a suitable change of the parametrization [TP 78] or by multiplication of the generating states with a suitable renormalization factor $r(a)$ [La 76], we can always achieve this property. The theory may also be developed more or less in the same way with weaker assumptions, in which there is a continuous spectrum ([TP 78]; see also Sec. 10.5).

and we can formally deduce the Hermitian eigenvalue problem

$$\mathcal{N}^{1/2}\mathcal{H}\mathcal{N}^{1/2}g = Eg \tag{10.11}$$

with

$$f = \mathcal{N}^{-1/2}g. \tag{10.12}$$

The inversion of $\mathcal{N}^{1/2}$ is impossible if $\mathcal{N}$ has zero eigenvalues ($n_k = 0$). Weight functions $f^0(a)$ which lie in the corresponding eigenspace produce many-body wave functions $|\Psi\rangle$ which vanish identically:

$$\langle\Psi|\Psi\rangle = \int da\, da'\, f^{0*}(a)\mathcal{N}(a, a')f^0(a') = 0. \tag{10.13}$$

If there exists a zero eigenvalue, the weight functions f corresponding to an HW state $|\Psi\rangle$ are not determined uniquely. We can always add a function f^0 and obtain the same $|\Psi\rangle$. To find a unique correspondence between $|\Psi\rangle$ and the weight function f, we should therefore at least restrict $f(a)$ to weight functions that are linear combinations of functions $u_k(a)$ with $n_k \neq 0$. For each of these functions, there exists a normalized vector in the Hilbert space

$$|k\rangle = \frac{1}{\sqrt{n_k}} \int u_k(a)|\Phi(a)\rangle\, da. \tag{10.14}$$

These vectors are called *natural states*. It is easy to show that they are orthogonal. They span a sub Hilbert space $\mathfrak{H}_C$, the so-called "collective" subspace. It is the smallest Hilbert space, which contains all the generating states $|\Phi(a)\rangle$. The projection onto this space is given by

$$P_C = \sum_{k,\, n_k \neq 0} |k\rangle\langle k|. \tag{10.15}$$

Instead of Eq. (10.11), we now diagonalize the Hamiltonian $P_C H P_C$ in the collective subspace

$$\sum_{k'} \langle k|H|k'\rangle g_{k'} = Eg_k \tag{10.16}$$

with

$$(P_C H P_C)_{kk'} = \langle k|H|k'\rangle = \int da\, da' \frac{u_k^*(a)}{\sqrt{n_k}} \mathcal{H}(a, a') \frac{u_{k'}(a')}{\sqrt{n_{k'}}}. \tag{10.17}$$

The eigenstates are given by

$$|\Psi\rangle = \sum_{k,\, n_k \neq 0} g_k |k\rangle. \tag{10.18}$$

From (10.14), we find the corresponding weight function

$$f(a) = \sum_{k,\, n_k \neq 0} \frac{g_k}{\sqrt{n_k}} u_k(a). \tag{10.19}$$

If there is only a finite number of eigenvalues $n_k \neq 0$ (or, more precisely, if zero is not a limit point of the n_ks), the function $f(a)$ is always well defined by (10.19). In the continuous case, however, there are often infinitely many *non*-vanishing eigenvalues n_k with zero as a limit point. It therefore happens that the functions $f(a)$ diverge, even if $|\Psi\rangle$ (10.18) has a finite norm ($\sum |g_k|^2 < \infty$), as we see from Eq. (10.19). This means that there are vectors in the Hilbert space $\mathfrak{H}_C$ that cannot be

represented as Hill–Wheeler states of the form (10.1) with well-behaved functions $f(a)$. The corresponding functions f have the character of distributions [see Eq. (10.58)]. It is obvious that we can never find such solutions of the Hill–Wheeler equation (10.5) by a discretization of the integration.*

Roughly speaking, the GCM ansatz (10.1) represents an expansion of the many-body wave functions $|\Psi\rangle$ in terms of the nonorthogonal generating functions $|\Phi(a)\rangle$: The expansion coefficients are given by the weight function $f(a)$: For a better understanding of the physical meaning of this function, it is useful to introduce the so-called *biorthogonal basis* [Wo 70]. It is given by the states $|\tilde{\Phi}(a)\rangle$, with the property

$$\langle\tilde{\Phi}(a)|\Phi(a')\rangle=\delta(a-a'). \tag{10.20}$$

If the norm has zero eigenvalues, we can obviously only require

$$\langle\tilde{\Phi}(a)|\Phi(a')\rangle=P_{\perp}(a,a')=\sum_{k,\,n_k\neq 0} u_k(a)u_k^*(a'), \tag{10.21}$$

where $P_{\perp}$ is the projection onto the space orthogonal to the eigenvectors $u_k(a)$ of Eq. (10.8) with eigenvalues $n_k=0$. We can construct vectors $|\tilde{\Phi}\rangle$ with this property by expanding them in natural states

$$|\tilde{\Phi}(a)\rangle=\sum_{k,\,n_k\neq 0}\frac{u_k^*(a)}{\sqrt{n_k}}|k\rangle. \tag{10.22}$$

The biorthogonal states $|\tilde{\Phi}(a)\rangle$ defined in this way, however, do not generally belong to the many-body Hilbert space, because their norm is not necessarily finite. Nevertheless, we can now express the weight function $f(a)$ belonging to the Hill–Wheeler state $|\Psi\rangle$ (10.1) in terms of

$$f(a)=\langle\tilde{\Phi}(a)|\Psi\rangle. \tag{10.23}$$

Therefore, $f(a)$ measures the overlap between the function $|\Psi\rangle$ and the state $|\tilde{\Phi}(a)\rangle$ in the biorthogonal basis. It is not the probability amplitude for finding the generating state $|\Phi(a)\rangle$ in the wave function $|\Psi\rangle$. The latter is given by $\langle\Phi(a)|\Psi\rangle$. The formula (10.23) again makes clear that $f(a)$ is not necessarily a well-behaved function.

So far, we treated only questions connected with the structure of the GCM representation (10.1). These properties were completely determined by the set of generating functions $|\Phi(a)\rangle$ used in this ansatz. In analogy to Chapter 9, we call them *kinematic properties*, because they have only to do with the underlying basis. We did not ask the question of whether the set $|\Phi(a)\rangle$ is useful for a description of the eigenstates of the Hamiltonian, that is, if $|\Psi\rangle$ can be represented at least in a good approximation as a linear superposition of these functions. This means that we did not investigate the *dynamic aspects* of the GCM ansatz.

We only can expect to obtain from Eq. (10.16) wave functions $|\Psi\rangle$ which are close to exact wave functions of the system, if the Hamiltonian

*The mathematical reason for these problems is the fact that the linear space formed by the Hill–Wheeler states (10.1) with mathematically well-behaved weight functions $f(a)$ is, in general, no Hilbert space, because it is not closed in the sense that for each converging set of vectors $|\Psi_i\rangle$ there exists a converging set of well-behaved weight functions $f_i(a)$.

commutes with P_C to a good approximation

$$[H, P_C] \simeq 0. \tag{10.24}$$

In such a case, the collective space contains eigenstates of H. To what degree this condition is fulfilled depends on the choice of the set $|\Phi(a)\rangle$.

10.2.3 Methods of Numerical Solution of the HW Equation

Based on the above consideration, we shall discuss very briefly several numerical methods for solving the Hill–Wheeler equations (10.5).

(i) The most obvious method is a *discretization of the integrals* in the variable a. As we have seen, in this case we end up with a matrix equation. We diagonalize the norm matrix and obtain an orthogonal set of eigenvectors. In a second step, we restrict ourselves to vectors with nonzero eigenvalues and, for numerical reasons, eigenvalues larger than a small positive constant ϵ, and then diagonalize the matrix (10.17). This gives energies E and wave functions of the form (10.18). In principle, we can also determine weight functions by Eq. (10.19). They are well defined for each choice of the discretization. It may happen, however, that they are ill behaved for the continuous problem. Choosing more and more mesh points for the discretization, we find in such cases convergence for the energy E and for the wave function $|\Psi\rangle$ (10.18), but no convergence for the weight function $f(a)$.

Small eigenvalues of the norm matrix are connected with approximate linear dependence of the generating functions. It is therefore not meaningful to increase the number of mesh points too much. Usually we limit the number of mesh points in such a way that there is no eigenvalue of the corresponding norm matrix smaller than a positive constant ϵ.

(ii) In some physical problems we use generating functions that allow an analytic solution of the eigenvalue problem (10.8), such that an *expansion in natural states* of the type (10.18) can be carried out. The approximation now consists of cutting this expansion after a finite number of natural states. Again we obtain a finite diagonalization problem (10.17), where the matrix elements can be calculated analytically.

In the special case where the norm $\mathcal{N}(a, a')$ only depends on the difference $(a - a')$, we can determine the eigenfunctions $u_k(a)$ immediately. They are given by plane waves

$$u_k(a) \propto e^{-ika} \tag{10.25}$$

and the corresponding eigenvalues are

$$n_k = \int da\, \mathcal{N}(a, 0) e^{ika}. \tag{10.26}$$

In this case the expansion in natural states corresponds to a Fourier transformation and the limitation to eigenvalues larger than ϵ excludes the components of the wave function with high k-values.

If the norm cannot be diagonalized analytically, it is often possible to approximate $\mathcal{N}(a, a')$ by a function $\mathcal{N}_0(a, a')$ which has this property. In such a case, we can carry out an expansion in natural states with respect to $\mathcal{N}_0$ and treat the difference $\mathcal{N} - \mathcal{N}_0$ by perturbation theory [LL 77].

(iii) A third type of approximation uses special properties of the overlap kernels $\mathcal{H}(a, a')$ and $\mathcal{N}(a, a')$. They are often sharply peaked at $a = a'$. Using this fact, we can transform the HW equation (10.5) into a *differential equation* of second order. We will discuss these methods in more detail in Section 10.7.

10.3 The Lipkin Model as an Example

In order to fix our ideas on a specific example we apply the foregoing considerations to the well-known Lipkin model (see Sec. 5.4).

In this model, from Eq. (5.48) we have for Eq. (10.2)

$$|a\rangle = N_0 e^{aK_+}|\Phi_0\rangle \qquad \langle a\,|\,a\rangle = 1 \tag{10.27}$$

and, with Eq. (E.51), obtain for the norm and Hamiltonian overlap

$$\mathcal{N}(a, b) = (1 + a^* b)^{\Omega}(1 + |a|^2)^{-\Omega/2}(1 + |b|^2)^{-\Omega/2} \tag{10.28}$$

and

$$\mathcal{H}(a, b) = -\frac{\epsilon\Omega}{2}\left\{\frac{1 - a^* b}{1 + a^* b} + \chi \cdot \frac{a^{*2} + b^2}{(1 + a^* b)^2}\right\}\mathcal{N}(a, b). \tag{10.29}$$

Decomposing the integral equation

$$\int d(\mathrm{Re}\, a')d(\mathrm{Im}\, a')(\mathcal{H}(a, a') - E\mathcal{N}(a, a'))f(a') = 0 \tag{10.30}$$

into its real and imaginary part we obtain two coupled two-dimensional integral equations for the real and imaginary part of $f(a')$. As we shall see, this equation contains the exact answer. It even contains the exact answer if we restrict ourselves to only real parameters $a = tg(\varphi/2)$. This yields

$$\int_{-\pi}^{\pi} d\varphi' \,[\mathcal{H}(\varphi, \varphi') - E\mathcal{N}(\varphi, \varphi')]f(\varphi') = 0, \tag{10.31}$$

where we use the same letter $\mathcal{H}$, $\mathcal{N}$, f for the transformed functions:

$$\mathcal{N}(\varphi, \varphi') = \left(\cos\left(\frac{\varphi - \varphi'}{2}\right)\right)^{\Omega},$$

$$\mathcal{H}(\varphi, \varphi') = -\frac{\epsilon\Omega}{2}\left\{\frac{\cos\Psi}{\cos(\phi/2)} + \frac{1}{2}\chi\left[\frac{1 + \sin^2\Psi}{\cos^2(\phi/2)} - 1\right]\right\}\mathcal{N}(\varphi, \varphi'), \tag{10.32}$$

with

$$\Psi = \tfrac{1}{2}(\varphi + \varphi'); \qquad \phi = \varphi - \varphi'. \tag{10.33}$$

In order to achieve the diagonization of the norm overlap analytically, we use the

identity $\cos(x)=\frac{1}{2}(e^{ix}+e^{-ix})$ and apply the binomial expansion:

$$\mathcal{N}(\varphi,\varphi')=\frac{1}{2^{\Omega}}\sum_{k=0}^{\Omega}\binom{\Omega}{k}e^{-ik\frac{1}{2}(\varphi-\varphi')}\cdot e^{i(\Omega-k)\frac{1}{2}(\varphi-\varphi')}$$

$$=\sum_{k=-\Omega/2}^{\Omega/2}e^{-ik\varphi}\frac{1}{2^{\Omega}}\binom{\Omega}{k+\frac{\Omega}{2}}e^{ik\varphi'}. \tag{10.34}$$

From the above form of the norm overlap, it becomes clear that it has the eigenfunctions $(1/\sqrt{2\pi})e^{-ik\varphi}$ with a finite number of non-vanishing eigenvalues

$$n_k=\frac{2\pi}{2^{\Omega}}\binom{\Omega}{k+\frac{\Omega}{2}}, \tag{10.35}$$

and that the Eq. (10.16) is a finite matrix equation

$$\sum_{k'}\tilde{H}_{kk'}g_{k'}=Eg_k, \tag{10.36}$$

with

$$\tilde{H}_{kk'}=\frac{1}{2\pi}\int_{-\pi}^{\pi}d\varphi\,d\varphi'\frac{e^{ik\varphi}}{\sqrt{n_k}}\mathcal{H}(\varphi,\varphi')\frac{e^{-ik'\varphi'}}{\sqrt{n_{k'}}}, \tag{10.37}$$

$$g_k=\sqrt{\frac{n_k}{2\pi}}\int_{-\pi}^{\pi}d\varphi\,e^{ik\varphi}f(\varphi). \tag{10.38}$$

The finiteness of the dimension in Eq. (10.36) clearly reflects the fact that in the Lipkin model containing Ω particles we cannot excite more than $\Omega+1$ different *ph* states going from $0p-0h$ up to the $\Omega p-\Omega h$ state. Since we started from a variational ansatz to arrive at Eq. (10.36), it is clear that the Hermitian eigenvalue problem (10.36) contains the exact answer for the Lipkin model.

It is very easy to evaluate the matrix elements (10.37); thus we have a straightforward method for solving the Lipkin model exactly. This shows us that, at least in this model, the GCM method is very powerful. We have to emphasize, however, that the structure of this model is extremely simple: There exists a collective subspace which coincides with the space spanned by the Hill–Wheeler representation (10.1). It is *finite* and, because of the quasi-spin symmetry, it *decouples exactly* from the rest of the Hilbert space. This is the optimal situation for the application of this method: In realistic cases we do not know a priori if similar conditions are fulfilled. Already in the case of the harmonic oscillator, which can still be treated analytically, but which has an infinite spectrum, we will run into some difficulties (see Sec. 10.5).

10.4 The Generator Coordinate Method and Boson Expansions

In the introduction we mentioned that the GCM is a method which is, in principle, capable of going beyond the boson expansion techniques. Nevertheless, in the limit of weak anharmonicities it should contain the Bose expansion theories. We will show in this section that this is actually the case [JDF 71, PUF 71].

We start from Eq. (10.2), that is, we use the most general product wave functions not orthogonal to $|\Phi_0\rangle$ as a set of generating functions. The formula of Onishi (E.51) and the quasi particle representation (E.18) gives us the overlap integrals for norm and Hamiltonian.*

$$\mathcal{N}(z,z'^*)=\langle\Phi(z^*)|\Phi(z'^*)\rangle=\exp\{\tfrac{1}{2}\mathrm{Tr}\ln(1-zz'^*)\},\quad(10.39)$$
$$\mathcal{H}(z,z'^*)=\langle\Phi(z^*)|\Phi(z'^*)\rangle=h(z,z'^*)\cdot\mathcal{N}(z,z'^*),$$

where

$$\begin{aligned}h(z,z'^*)=&\sum_k E_k\rho_{kk}+3\sum_{klpq}\left(H^{40}_{klpq}\bar\kappa_{kl}\bar\kappa_{pq}+H^{40*}_{klpq}\kappa_{kl}\kappa_{pq}\right)\\&+3\sum_{klpq}\left(H^{31}_{klpq}\bar\kappa_{kl}\rho_{qp}+H^{31*}_{klpq}\rho_{pq}\kappa_{kl}\right)\\&+\frac{1}{4}\sum_{klpq}\left(H^{22}_{klpq}\bar\kappa_{kl}\kappa_{pq}+2H^{22}_{klpq}\rho_{pk}\rho_{ql}\right)\end{aligned}\quad(10.40)$$

and

$$\bar\kappa_{kl}\mathcal{N}(z,z'^*)=\left\{z\frac{1}{1-z'^*z}\right\}_{kl}\mathcal{N}(z,z'^*)=\langle\Phi(z^*)|\alpha_k^+\alpha_l^+|\Phi(z'^*)\rangle,\quad(10.41)$$

$$\kappa_{kl}\mathcal{N}(z,z'^*)=\left\{\frac{1}{1-z'^*z}z'^*\right\}_{kl}\mathcal{N}(z,z'^*)=\langle\Phi(z^*)|\alpha_l\alpha_k|\Phi(z'^*)\rangle,\quad(10.42)$$

$$\rho_{kl}\mathcal{N}(z,z'^*)=-\left\{z'^*z\frac{1}{1-z'^*z}\right\}_{kl}\mathcal{N}(z,z'^*)=\langle\Phi(z^*)|\alpha_l^+\alpha_k|\Phi(z'^*)\rangle.\quad(10.43)$$

If we use z_{kl} and z^*_{kl} $(k<l)$ as independent variables in the sense of Section 5.2 we obtain the relation $(k<l, k'<l')$

$$\left[\frac{\partial}{\partial z_{kl}}, z_{k'l'}\right]=\delta_{ll'}\delta_{kk'},\qquad\left[\frac{\partial}{\partial z_{kl}}, z^*_{k'l'}\right]=0.$$

Then the quantities ρ_{kl}, κ_{kl}, $\bar\kappa_{kl}$, and therefore h, can be expressed as operators acting on $\mathcal{N}(z,z'^*)$:

$$\begin{aligned}\bar\kappa_{kl}\mathcal{N}(z,z'^*)&=\left(z_{kl}-\sum_{pq}z_{kp}z_{lq}\frac{\partial}{\partial z_{pq}}\right)\mathcal{N}(z,z'^*)\equiv\hat{\bar\kappa}_{kl}\mathcal{N}(z,z'^*),\\\kappa_{kl}\mathcal{N}(z,z'^*)&=\frac{\partial}{\partial z_{kl}}\mathcal{N}(z,z'^*)\equiv\hat\kappa_{kl}\mathcal{N}(z,z'^*),\\\rho_{kl}\mathcal{N}(z,z'^*)&=\sum_p z_{lp}\frac{\partial}{\partial z_{kp}}\equiv\hat\rho_{kl}\mathcal{N}(z,z'^*).\end{aligned}\quad(10.44)$$

The relations (10.44) are somewhat tedious but straightforward to verify in differentiating the norm overlap. For the reduced Hamiltonian (10.40), Eq. (10.44) means that the density operators κ, $\bar\kappa$ and ρ can be replaced by

*The definition of the overlap functions $\mathcal{N}(z,z')$, $\mathcal{H}(z,z')$, $h(z,z')$ in this subsection deviates from Eq. (10.7) for complex values of z. It makes it clear that these functions depend actually on z and not on z^* (see also Sec. 10.6).

differential operators. In a shorthand notation we therefore get

$$\mathcal{H}(z,z'^*)=h(z,z'^*)\mathcal{N}(z,z'^*)\equiv\tilde{h}\left(z,\frac{\partial}{\partial z}\right)\mathcal{N}(z,z'^*). \tag{10.45}$$

With Eq. (10.5) and the definition

$$g(z)=\int dz' f(z')\mathcal{N}(z,z'^*), \tag{10.46}$$

we obtain the Schrödinger equation

$$\tilde{h}\left(z,\frac{\partial}{\partial z}\right)g(z)=Eg(z). \tag{10.47}$$

It turns out that the operator $\tilde{h}$ in Eq. (10.47) and the boson Hamiltonian (9.93) of the Dyson expansion are identical if we identify the operators z_{kl} and $\partial/\partial z_{kl}$ with the boson operators B_{kl}^{+}, B_{kl}, respectively [JDF 71]. z and $\partial/\partial z$ are correspondingly the Bargmann representation of B^{+} and B [Ba 62, JS 64]. At this point, this step is completely formal. Its mathematical background is discussed in Section 10.6.

Since we showed in Section 9.2.6 that, to infinite order, all boson expansions are equivalent and that they can be mutually transformed into one another, it is not astonishing that the Marumori and Belyaev–Zelevinskii boson expansions can also be derived with the GCM [Ho 72]. For instance, the RPA was deduced in this way in an early paper by Jancovici and Schiff [JS 64].*

We know from Chapter 9 that we can represent the exact eigenfunction of the Schrödinger equation within the boson space. It is therefore now clear that the ansatz (10.2) contains the exact solution not only in the Lipkin model (Sec. 10.3) but also in the general case. Contrary to the Lipkin model, however, the representation (10.2) is not necessarily unique in the general case, because there is no unique solution f of Eq. (10.47) if the integral operator $\mathcal{N}(z,z')$ has zero eigenvalues. In fact, this is usually the case, because $\mathcal{N}(z,z')$ is totally antisymmetric with respect to the indices i_1k_1, $i_2k_2,\ldots$ in the arguments $z'_{i_1k_1}, z'_{i_2k_2}$ and projects all symmetric parts of the function $f(z')$ onto zero. To get a unique description, we shall therefore restrict ourselves to functions f, which are totally antisymmetric.

To solve the Hill–Wheeler equations for all generator coordinates z_{ik} exactly is certainly impossible. In practice, we are restricted to one or perhaps a few coordinates. It is a problem in itself to determine the optimal generator coordinates and wave functions. We will come back to this in Section 10.8.

In most cases we rely on phenomenology for the choice of a few collective parameters $a_1, a_2,\ldots$ as generator coordinates. Then the set of variational wave functions (10.1) contains only a small part of the many-body Hilbert space, but we hope that we are able to at least describe the collective modes a nucleus may undergo. It is clear that this is only

* See also [BW 68].

possible if the collective motion is well separated from all other degrees of freedom that are not taken into account by the ansatz (10.1).

10.5 The One-Dimensional Harmonic Oscillator

A model which is often used to show the problems that can arise in the GCM is the one-dimensional harmonic oscillator. It is also of great importance in the theory of collective motion, since we have seen that the collective excitations at low energy can be described quite well in the harmonic approximation (Chap. 8). Assuming only one collective degree of freedom, we get:

$$H=\frac{1}{2M}\hat{P}^2+\frac{M\omega^2}{2}\hat{Q}^2. \tag{10.48}$$

We use *generating functions* of Gaussian form in the q-representation ($\hat{Q}|q\rangle = q|q\rangle$):

$$\langle q|\Phi(a)\rangle=(\pi s^2)^{-1/4}e^{-(q-a)^2/2s^2}. \tag{10.49}$$

and obtain for the norm and Hamilton *overlap functions*

$$\mathcal{N}(a,a')=e^{-(a-a')^2/4s^2}, \tag{10.50}$$

$$\mathcal{H}(a,a')=\mathcal{N}(a,a')\left\{\frac{\hbar^2}{2M}\left(\frac{1}{2s^2}-\frac{(a-a')^2}{4s^4}\right)+\frac{M\omega^2}{2}\left(\frac{s^2}{2}+\frac{(a+a')^2}{4}\right)\right\}. \tag{10.51}$$

An analytical solution of the corresponding Hill–Wheeler equation (10.5) can be given [GW 57]. But to construct the collective space $\mathfrak{H}_C$, we shall apply in the following the method (ii) described in Section 10.2.3.

The norm overlap has a Gaussian form and depends only on the difference $(a-a')$. As discussed in Section 10.2.3, it is diagonalized by the functions $u(k,a)$ with eigenvalues $n(k)$*

$$u(k,a)=\frac{1}{\sqrt{2\pi}}e^{-ika}, \qquad n(k)=\int_{-\infty}^{\infty}da\,e^{ika}\mathcal{N}(a,0)=2\sqrt{\pi s^2}\,e^{-k^2s^2}, \tag{10.52}$$

where k is now a continuous index. As we see, the spectrum has no zero eigenvalues for finite k's, but 0 is a limit point for $k\to\infty$. The representation of the Hamiltonian in this basis is

$$H_{kk'}=\frac{\hbar k^2}{2M}\delta(k-k')-\frac{M\omega^2}{2}\delta''(k-k'), \tag{10.53}$$

which is obviously the momentum space version of the standard harmonic oscillator. The important point is that the parameter s, which depends on the choice of the generator wave functions, has disappeared. This is the exact Hamiltonian, and we gain all the exact solutions from it.

The *Schrödinger equation* in the collective subspace is

$$\frac{\hbar^2k^2}{2M}g(k)-\frac{M\omega^2}{2}\frac{d^2}{dk^2}g(k)=Eg(k). \tag{10.54}$$

*The operator $\mathcal{N}$ in Eq. (10.50) is not of Hilbert–Schmidt type. It therefore has no discrete spectrum.

The solution for the ground state is

$$g_0(k)=\left(\frac{b^2}{\pi}\right)^{1/4} e^{-\frac{1}{2}b^2k^2}, \tag{10.55}$$

with

$$b=\sqrt{\frac{\hbar}{M\omega}}\ .$$

The corresponding weight function $f_0(a)$ can be found from Eqs. (10.19) and (10.52) or by an explicit solution of the HW equation (10.6). For $b>s$ we get:

$$f_0(a)=\int dk\ g_0(k)\frac{1}{\sqrt{n(k)}}u(k,a)=\frac{1}{\sqrt{2\pi}}\frac{1}{\sqrt{b^2-s^2}}\sqrt{\frac{b}{s}}\cdot e^{-a^2/2(b^2-s^2)}. \tag{10.56}$$

The *ground state wave function* $|\Psi_0\rangle$ in Q space is then given by

$$\Psi_0(q)=\int da\ f_0(a)\langle q|\Phi(a)\rangle=(\pi\cdot b^2)^{-1/4}e^{-q^2/2b^2}. \tag{10.57}$$

We have thereby found the correct ground state wave function of the harmonic oscillator. The functions f_i can also be explicitly constructed, yielding the excited states. In Eq. (10.57) we can see that the weight function f_0 has added to the width s of the generating function $|\Phi(a)\rangle$ just what was lacking to generate the width b of the harmonic oscillator ground state. However, this is only possible if the width of the generating function is smaller than that of the solution, since by a convolution the width can only increase. Eq. (10.56) is therefore only valid if $b>s$.

If, on the contrary, $b<s$, $f_0(a)$ is no longer given by an ordinary function, but by a distribution which is necessary in order to "sharpen" the generating function. However, we can derive $f_0(a)$ from the integral (10.56) by expanding the integrand in powers of (s^2-b^2); then:

$$f_0(a)=\sqrt{\frac{b}{s}}\sum_{n=0}^{\infty}\left(\frac{1}{2}(s^2-b^2)\right)^n\frac{(-)^n}{n!}\delta^{(2n)}(a) \tag{10.58}$$

is a solution of the HW equation for $s>b$. It is obvious that we never could have obtained this solution by discretizing the integral equation.

This example shows very nicely that for $s>b$ it is not possible to express the exact ground state with a well-behaved weight function f_0. The collective space $\mathfrak{H}_C$ is, in this case, the whole Hilbert space. We have to emphasize, however, that a *numerical solution* with discrete mesh points converges to the right groundstate wave function $\Psi_0(q)$ (10.57) for all values of the width s (this can be shown from the general Tauberian theorem [Wi 33]). Only the weight function $f_0(a)$ remains undefined.

There is another way to avoid this difficulty. We can use the analytical behavior of $f(a)$ in choosing *complex* a. For example, we obtain the solution (10.57) for $s>b$ with the analytical $f(a)$ of Eq. (10.56) in integrating along the pure imaginary a-axis ($a=a_r+ia_i$):

$$\Psi_0(q)=\int_{-\infty}^{+\infty}da_i\ f_0(ia_i)\langle q|\Phi(ia_i)\rangle=(\pi b^2)^{-1/4}e^{-q^2/2b^2}. \tag{10.59}$$

Here we have only used the fact that in giving the generating function for real parameters we necessarily also have its analytic continuation into the complex plane. We then have to deform the path of integration in such a way as to enter a

domain where the weight function $f_0(a)$ is well behaved. For a numerical solution, however, this does not help very much, because in order to find the "good" integration path we already have to know the solution.

A way out of this problem is to introduce *complex generator coordinates* and integrate over the whole complex plane. We will see in the next section that this yields very well-behaved weight functions in the case of a Gaussian overlap. Introducing the complex space, however, renders the problem numerically much more laborious, since this means, as we have seen in Eq. (10.3), the introduction of a two-parameter GCM. On the other hand, it might not help to avoid ill-behaved weight functions f in the general case.

10.6 Complex Generator Coordinates

A formulation of quantum mechanics exists based on operators with a complex spectrum—the so-called *Bargmann representation* [Ba 62]. The corresponding basis functions are not orthogonal in the usual sense. The corresponding Schrödinger equation is therefore of the Hill–Wheeler type (10.5) with complex generator coordinates. The method of complex generator coordinates in the general sense is closely related to this representation of quantum mechanics and we will discuss its properties in this section. For the sake of simplicity we restrict our discussion to one dimension.

10.6.1 The Bargmann Space

The states of a quantum mechanical system $|\varphi\rangle$ are vectors in an abstract Hilbert space. For concrete calculation we need some representation, for instance, the wave functions in coordinate space

$$\varphi(q)=\langle q|\varphi\rangle, \tag{10.60}$$

that is, we use the eigenfunctions $|q\rangle$ of the self-adjoint operator $\hat{Q}$ as a basis. We can also use the momentum operator $\hat{P}$ and then have the representation in a Fourier space. The eigenvalues of these operators q or p are real, therefore we can work with real coordinates q or p. This is an advantage of such representations. A disadvantage is the fact that the eigenfunction $|q\rangle$ or $|p\rangle$ has an infinite norm. They can only be normalized to a δ function

$$\langle q|q'\rangle=\delta(q-q'); \qquad \langle p|p'\rangle=\delta(p-p'). \tag{10.61}$$

Instead of $\hat{Q}$ and $\hat{P}$, Fock [Fo 28] introduced the operators

$$O=\sqrt{\frac{M\omega}{2\hbar}}\left(\hat{Q}+\frac{i}{M\omega}\hat{P}\right); \qquad O^+=\sqrt{\frac{M\omega}{2\hbar}}\left(\hat{Q}-\frac{i}{M\omega}\hat{P}\right), \tag{10.62}$$

where the constants ω and M are arbitrary at this point. They obey the commutation relation

$$[O, O^+]=1. \tag{10.63}$$

Bargmann [Ba 62] used the operator O to derive a new representation. The non-Hermitian operator O can be diagonalized and has all the complex numbers z as its eigenvalues

$$O|z\rangle = z|z\rangle. \tag{10.64}$$

The eigenfunctions are called *coherent states* [Gl 63] because they are of great use in the description of coherent phenomena of light.* They can be represented by the eigenstates $|n\rangle$ of the operator $\hat{N} = O^+O$ (which are oscillator functions)

$$\hat{N}|n\rangle = n|n\rangle \tag{10.65}$$

in the following way†

$$|z\rangle = \sum_{n=0}^{\infty} \frac{z^n}{\sqrt{n!}}|n\rangle = e^{zO^+}|0\rangle = e^{\frac{1}{2}|z|^2} e^{(zO^+ - z^*O)}|0\rangle. \tag{10.66}$$

$|0\rangle$ is the ground state of the harmonic oscillator. It is easy to derive this relation from the property $O|n\rangle = \sqrt{n}\,|n-1\rangle$ [see Eq. (C.11)]. Similarly, we get

$$O^+|z\rangle = \frac{d}{dz}|z\rangle. \tag{10.67}$$

The orthogonality relation (10.61) now has the form

$$\langle z|z'\rangle = \exp(z^*z') := B(z^*, z'), \tag{10.68}$$

which means that these states have a finite norm, but are no longer orthogonal in the usual sense. However, it is possible to define a real positive *measure* $d\mu(z)$ in the complex plane.

$$d\mu(z) = \frac{1}{\pi} e^{-|z|^2}\, dx\, dy, \qquad \text{where } z = x + iy, \tag{10.69}$$

which allows us to derive the closure relation

$$1 = \sum_n |n\rangle\langle n| = \int |z\rangle\, d\mu(z)\, \langle z|, \tag{10.70}$$

where the integral runs over the whole complex plane. Equation (10.70) can be proved with Eqs. (10.66) and (10.69) using polar coordinates $z = re^{i\varphi}$.

In analogy to Eq. (10.60), the wave functions in this representation are given by

$$\psi(z) = \langle z^*|\psi\rangle. \tag{10.71}$$

From (10.66) we see that they depend on z, not on z^*. The measure (10.69) allows us to define a scalar product

$$(\psi_1|\psi_2) = \int d\mu(z)\, \psi_1^*(z)\psi_2(z) = \langle\psi_1|\psi_2\rangle, \tag{10.72}$$

* For a general group theoretical definition of coherent states, see [Pe 77].

† We use the Baker–Campbell–Hausdorff formula [Wi 67]

$$e^A \cdot e^B = e^{A+B+\frac{1}{2}[A,B]}$$

for operators A and B, that commute with $[A, B]$.

which means that the representation (10.71) is isometric with respect to the measure (10.69).We can now also derive a representation for the operators O, O^+ from (10.64) and (10.72):

$$O^+\psi(z)=\langle z^*|O^+|\psi\rangle=z\langle z^*|\psi\rangle=z\psi(z),$$

$$O\psi(z)=\langle z^*|O|\psi\rangle=\frac{d}{dz}\langle z^*|\psi\rangle=\frac{d}{dz}\psi(z), \tag{10.73}$$

or

$$O^+=z, \qquad O=\frac{d}{dz}, \tag{10.74}$$

which obviously fulfills the commutation relation (10.63).*

Finally, with Eq. (10.70) it is easy to prove that the function $B(z^*, z')$ in Eq. (10.68) has with the metric (10.72) similar properties in the Bargmann space as the δ function in the usual space, namely

$$\psi(z)=\int B(z,z'^*)\psi(z')\,d\mu(z'). \tag{10.75}$$

This function is therefore often called Bargmann's complex δ function. In fact, it is an entire function and no distribution.

10.6.2 The Schrödinger Equation

We obtain the Schrödinger equation $H|\psi\rangle=E|\psi\rangle$ as represented in the Bargmann space by multiplying by $\langle z^*|$ and inserting the complete set (10.70)

$$\int H(z,z'^*)\psi(z')d\mu\,(z')=E\psi(z), \tag{10.76}$$

where

$$H(z,z'^*)=\langle z^*|H|z'^*\rangle \tag{10.77}$$

is a function of z, z'^*. It does not depend on z^* or z'.

We can rewrite Eq. (10.76) in two ways [UB 68]:

(i) We can use Eq. (10.75) and gain an equation of the Hill–Wheeler form

$$\int(H(z,z'^*)-EB(z,z'^*))\psi(z')\,d\mu(z')=0. \tag{10.78}$$

This corresponds to the GCM ansatz

$$|\psi\rangle=\int d\mu\,(z)\psi(z)|z^*\rangle, \tag{10.79}$$

where the coherent states $|z^*\rangle$ of Eq. (10.66) have been used as generating functions with the complex generator coordinate z. The

*One should not be confused by the fact that (10.74) is different from (10.64) and (10.67). The reason is that there are two different representations of O^+, O acting in different Hilbert spaces.

advantage of these generating functions is that their overlap $B(z, z'^*)$ corresponds to the δ function in the Bargmann space. It has the more or less Gaussian form

$$B(z, z'^*) = e^{zz'^*} = e^{\frac{1}{2}z^2} e^{\frac{1}{2}z'^{*2}} e^{-\frac{1}{2}(z-z'^*)^2}. \tag{10.80}$$

(ii) We can introduce the function $h(z, z'^*)$

$$H(z, z'^*) = h(z, z'^*) B(z, z'^*) \tag{10.81}$$

and use

$$\frac{\partial}{\partial z} B(z, z'^*) = z'^* B(z, z'^*) \tag{10.82}$$

to transform to the differential equation

$$: h\left(z, \frac{\partial}{\partial z}\right) : \psi(z) = E\psi(z). \tag{10.83}$$

The dots mean normal ordering so that powers of z are always on the left side of $\partial/\partial z$. If we now use the operators (10.74) we can express the Hamiltonian H by boson operators and get a "boson expansion," as discussed in Section 10.4. There, however, we do not have coherent states because of the non-Gaussian form of the norm (10.39). Nevertheless, this may often be quite a good approximation (see Sec. 10.7).

Because of the δ function property of the function $B(z, z'^*)$ in Eq. (10.75) the function g of Eq. (10.12) now coincides with the weighting function, that is, the norm overlap is in this case just the identity operator. Therefore, the weight functions are now always well behaved, which was not true in the case of real generator coordinates. However, we have to emphasize that this holds only for overlap functions of a Gaussian form. In the general case the use of complex coordinates does not automatically yield well-behaved weight functions.

10.6.3 Gaussian Wave Packets in the Harmonic Oscillator

Gaussian wave packets in the harmonic oscillator are closely related to the theory of coherent states. Therefore, in the following we discuss some of their properties and, particularly, their time dependence. However, we do not need all of the following results in the context of the generator coordinate method, in which we treat only the stationary problem.

The coherent state $|z\rangle$ in Eq. (10.66) is represented by the boson operators O^+ and O of a harmonic oscillator Hamiltonian.

$$H_0 = \frac{\hat{P}^2}{2M} + \frac{M\omega^2}{2} \hat{Q}^2 = \hbar\omega(O^+ O + \tfrac{1}{2}),$$

with the oscillator length $b = (\hbar/M\omega)^{1/2}$. The state $|z\rangle$ in Eq. (10.66) is not

normalized. In the following we therefore define normalized states:

$$|z,0\rangle = e^{-\frac{1}{2}|z|^2}|z\rangle = \exp(zO^+ - z^*O)|0\rangle,$$

where $|0\rangle$ corresponds to the ground state of H_0. The unitary operator

$$\hat{S}_z^+ = \exp(zO^+ - z^*O)$$

produces a shift in the boson operators by an amount z (see Sec. E.5)

$$\hat{S}_z^+ O^+ \hat{S}_z = O^+ - z^*, \qquad \hat{S}_z^+ O \hat{S}_z = O - z.$$

Decomposing z into its real and imaginary parts

$$z = \frac{1}{\sqrt{2}} \frac{1}{b}\left(q_0 + i\frac{p_0}{M\omega}\right),$$

we can also express it by the operators $\hat{P}$ and $\hat{Q}$,

$$\hat{S}_z^+ = \exp\frac{i}{\hbar}(p_0\hat{Q} - q_0\hat{P}) = \exp\left(-\frac{i}{2\hbar}p_0q_0\right)\exp\left(\frac{i}{\hbar}p_0\hat{Q}\right)\cdot\exp\left(-\frac{i}{\hbar}q_0\hat{P}\right),$$

which shows that it produces a shift in the coordinate by an amount q_0 and in the momentum by an amount p_0,

$$\hat{S}_z^+ \hat{Q} \hat{S}_z = \hat{Q} - q_0, \qquad \hat{S}_z^+ \hat{P} \hat{S}_z = \hat{P} - p_0.$$

The coherent state

$$|z,0\rangle = \exp\frac{i}{\hbar}(p_0\hat{Q} - q_0\hat{P})|0\rangle \tag{10.84}$$

is therefore nothing but the ground state of a shifted harmonic oscillator

$$H_z = \frac{(\hat{P}-p_0)^2}{2M} + \frac{M\omega^2}{2}(\hat{Q}-q_0)^2. \tag{10.85}$$

It is neither an eigenstate of $\hat{P}$ nor $\hat{Q}$ nor of the original Hamiltonian H_0. We obtain for the mean value of the coordinate and the momentum operator

$$\langle z,0|\hat{Q}|z,0\rangle = q_0, \qquad \langle z,0|\hat{P}|z,0\rangle = p_0,$$

and for the fluctuations

$$\langle z,0|\Delta\hat{Q}^2|z,0\rangle = \frac{b^2}{2}, \qquad \langle z,0|\Delta\hat{P}^2|z,0\rangle = \frac{\hbar^2}{2b^2}. \tag{10.86}$$

This shows that $|z,0\rangle$ minimizes the uncertainty

$$\langle z,0|\Delta\hat{Q}^2|z,0\rangle\langle z,0|\Delta\hat{P}^2|z,0\rangle = \frac{\hbar^2}{4}.$$

Therefore, the coherent states are the "most classical" wave functions that can be found.

We can represent the state $|z,0\rangle$ in coordinate space by the wave function

$$\psi_0(x) = \langle x|z,0\rangle = e^{-\frac{1}{2}|z|^2}\langle x|e^{zO^+}|0\rangle.$$

Using $O^+ = O - i\sqrt{2}\, b\hat{P}/\hbar$ and $O|0\rangle = 0$, we obtain

$$\psi_0(x) = e^{-\frac{1}{2}(|z|^2 - z^2)} \langle x - \sqrt{2}\, bz|0\rangle.$$

This is a Gaussian wave function centered around $\sqrt{2}\, bz = q_0 + i(p_0/M\omega)$

$$\psi_0(x) = (\sqrt{\pi}\, b)^{-1/2} e^{-\frac{1}{2}(|z|^2 - z^2)} e^{-(x/\sqrt{2}b - z)^2}$$

$$= (\sqrt{\pi}\, b)^{-1/2} \exp \frac{i}{\hbar} \left\{ \frac{p_0 q_0}{2} + p_0(x - q_0) + i\frac{M\omega}{2}(x - q_0)^2 \right\}. \quad (10.87)$$

Since this wave function is not an eigenfunction of the original Hamiltonian H_0, the coherent state $|z, 0\rangle$ forms a wave packet. We obtain its time evolution by

$$|z, t, 0\rangle = e^{-\frac{i}{\hbar} H_0 t} |z, 0\rangle = e^{-\frac{i\omega}{2} t} |ze^{-i\omega t}, 0\rangle.$$

Decomposing $ze^{-i\omega t}$ into real and imaginary parts, we find

$$ze^{-i\omega t} = \frac{1}{\sqrt{2}\, b} \left(q(t) + i\frac{p(t)}{M\omega} \right),$$

where

$$q(t) = q_0 \cos \omega t + \frac{p_0}{M\omega} \sin \omega t,$$

$$p(t) = p_0 \cos \omega t - M\omega q_0 \sin \omega t$$

describes the classical path of a particle with the initial conditions q_0, p_0 at $t = 0$. To obtain the time-dependent wave function in coordinate space, we have only to replace p_0 and q_0 in Eq. (10.87) by $p(t)$ and $q(t)$, and to add the trivial phase $-\omega t/2$:

$$\psi_0(x, t) = (\sqrt{\pi}\, b)^{-1/2} \exp \frac{i}{\hbar} \{ S(x, t) - \epsilon_0 t \},$$

with

$$S(x, t) = W(t) + p(t)(x - q(t)) + i\frac{M\omega}{2}(x - q(t))^2, \quad (10.88)$$

$$W(t) = \frac{1}{2} p(t) q(t), \qquad \epsilon_0 = \frac{\hbar\omega}{2}.$$

It describes a Gaussian wave packet whose center $q(t)$ moves like a classical particle: It oscillates with constant frequency ω. The wave packet itself preserves its shape and shows no dispersion, that is, it moves "coherently."

We obtained this wave packet as the time evolution of the stationary ground state of a shifted harmonic oscillator Hamiltonian. We can also take internal excitations of this wave packet into account by starting with the excited states in the shifted oscillator (10.85)

$$|z, n\rangle = \hat{S}_z^+ |n\rangle = \frac{1}{\sqrt{n!}} (O^+ - z^*)^n |z, 0\rangle.$$

Again, their time dependence is given by

$$|z,t,n\rangle = e^{-\frac{i}{\hbar}Ht}|z,n\rangle = e^{-\frac{i}{\hbar}\epsilon_n t}|ze^{-i\omega t},n\rangle,$$

with $\epsilon_n = \hbar\omega(n+\frac{1}{2})$.

For the corresponding wave function in the coordinate space a lengthy calculation gives

$$\psi_n(x,t) = \langle x|z,t,n\rangle = (2^n n!\sqrt{\pi}\, b)^{-1/2} H_n\left(\frac{x-q(t)}{b}\right)\exp\frac{i}{\hbar}(S(x,t)-\epsilon_n t),$$

where H_n are the Hermite polynomials and $S(x,t)$ is given in Eq. (10.88). Again, these Hermite wave packets $\psi_n(x,t)$ oscillate along the classical path $q(t)$, preserving their shape. At each time they represent a complete orthogonal set, which may be an efficient tool for the investigation of time-dependent processes. So far we started with stationary wave functions in a shifted oscillator (10.85) having the same frequency ω, that is, the same oscillator length b as the original Hamiltonian H_0. We can go a step further still and start with the eigenstates in a harmonic oscillator potential

$$H_{\bar{z}} = \frac{(\hat{P}-p_0)^2}{2M} + \frac{M\bar{\omega}^2}{2}(\hat{Q}-q_0)^2$$

with a different frequency $\bar{\omega}\neq\omega$ and the corresponding oscillator length $\bar{b}$. Its eigenstates are given by

$$|\overline{\bar{z},n}\rangle = \frac{1}{\sqrt{n!}}(\bar{O}^+ - \bar{z}^*)^n S_{\bar{z}}^+|\bar{0}\rangle,$$

with

$$\bar{O}^+ = \frac{1}{\sqrt{2}}\frac{1}{\bar{b}}\left(\hat{Q} - \frac{i}{M\bar{\omega}}\hat{P}\right) \quad \text{and} \quad \bar{O}|\bar{0}\rangle = 0.$$

Without proof, we give the coordinate representation of the time-dependent wave function (for details, see [Bö 76, LR 69]):

$$\bar{\psi}_n(x,t) = \langle x|e^{-\frac{i}{\hbar}H_0 t}|\overline{\bar{z},n}\rangle$$

$$= \left(2^n n!\sqrt{\pi}\,\bar{b}(t)\right)^{-1/2} H_n\left(\frac{x-q(t)}{\bar{b}(t)}\right)\exp\left[-\frac{1}{2}\frac{(x-q(t))^2}{\bar{b}(t)^2} + \frac{i}{\hbar}\phi(x,t)\right], \tag{10.89a}$$

with

$$\bar{b}(t) = b(\sigma+\tau\cos 2\omega t)^{1/2}$$

$$\phi(x,t) = W(t) + p(t)(x-q(t)) - \frac{M\omega}{2}(x-q(t))^2\frac{\tau\sin 2\omega t}{\sigma+\tau\cos 2\omega t}$$

$$-\hbar\left(n+\frac{1}{2}\right)\operatorname{arctg}\left(\frac{\bar{\omega}}{\omega}\operatorname{tg}(\omega t)\right) \tag{10.89b}$$

and

$$\sigma = \frac{1}{2}\left(\frac{\omega}{\bar{\omega}} + \frac{\bar{\omega}}{\omega}\right), \qquad \tau = \frac{1}{2}\left(\frac{\omega}{\bar{\omega}} - \frac{\bar{\omega}}{\omega}\right). \tag{10.89c}$$

10.6.4 Double Projection

In Section 10.5 we treated the case of a harmonic oscillator using generating wave functions, which were shifted in the configuration space by the amount a (10.49). Up to a factor $e^{1/2|z|^2}$, these are coherent states of the form

$$|z\rangle = e^{\frac{1}{2}|z|^2} \exp\left\{\frac{i}{\hbar}(\beta\hat{Q} - \alpha\hat{P})\right\}|0\rangle,$$

$$\alpha = \sqrt{\frac{2\hbar}{M\omega}}\ \mathrm{Re}\, z, \qquad \beta = \hbar\sqrt{\frac{2M\omega}{\hbar}}\ \mathrm{Im}\, z,$$

with a real parameter z: $\alpha = a$, $\beta = 0$. Using complex values of z means an analytic continuation of these functions into the complex plane or the introduction of a second generator coordinate $\beta \propto \dot{\alpha}$, which describes a shift in momentum.

Such a treatment—using generating functions $|\Phi(\alpha, \beta)\rangle$ depending not only on the coordinate α, but also on the corresponding momentum β—was first introduced by Thouless and Peierls [PT 62] in connection with projection onto eigenstates of linear momentum, and is therefore called the double projection technique (see Chap. 11). In the case of generating functions of the oscillator type, this method is useful insofar as it is equivalent to the use of coherent states as generating functions that are very well behaved in the Bargmann space. We do not run into trouble, even if their width s is larger than the width of the function $|\Psi\rangle$ we wish to represent (see Sec. 10.5).

However, as we have seen in Section 10.5, we already get the exact wave function with one real generator coordinate α if we construct the collective space $\mathfrak{H}_C$ in a proper way. Only the weight functions are very ill behaved in some cases. This means that the collective space $\mathfrak{H}_C$ spanned by the generating functions $|\Phi(\alpha, 0)\rangle$ is the same as the space $\mathfrak{H}_C$ spanned by the functions $|\Phi(\alpha, \beta)\rangle$ in the harmonic oscillator case. The coordinate β is *redundant*.

In general, this may not always be the case. We might sometimes be tempted to introduce a two-parameter generator coordinate for physical reasons. For example, for the case of quadrupole oscillations, one might think that not only the quadrupole moment α is a good generator coordinate, but in addition also its time derivative $\dot{\alpha}$ (or the corresponding momentum β), thus forming a set of generator coordinates of two conjugate variables. We will see in Chapter 12 that in the time-dependent Hartree–Fock (TDHF) theory, we always have to deal with such wave

functions $|\Phi(\alpha, \beta)\rangle$. In the GCM, however, it often happens that the generating functions $|\Phi(\alpha, \beta)\rangle$ lie mostly in the space already formed by the $|\Phi(\alpha, 0)\rangle$. They are, therefore, almost redundant. The determination of the weight function $f(\alpha, \beta)$ through the variational principle becomes very insensitive in such cases, or, in other words, the norm overlap $\mathcal{N}(\alpha\beta, \alpha'\beta')$ has a large eigenspace with vanishing norm.

The two parameters (α, β) are certainly redundant, if they only enter into the combination $\alpha + i\lambda\beta$, where λ is a real constant. From the relation

$$|\Phi(\alpha + i\lambda\beta)\rangle = \exp\left(i\lambda\beta\frac{\partial}{\partial\alpha}\right)|\Phi(\alpha, \beta)\rangle$$

and subsequent partial integration in the HW-ansatz (10.1), we can redefine new weight functions depending on only one real parameter.

A function $|\Phi(\alpha, \beta)\rangle$ only depends on the combination $\alpha + i\lambda\beta$ if

$$\left(\frac{\partial}{\partial\beta} - i\lambda\frac{\partial}{\partial\alpha}\right)|\Phi(\alpha, \beta)\rangle = 0. \tag{10.90}$$

This condition means local redundancy [Re 76a+b]: The change of the function $|\Phi\rangle$ by an infinitesimal variation of the coordinate β can be expressed by a corresponding change of the coordinate α. It is a sufficient condition for redundancy.

The double projection method in the GCM theory has not, up to the present, been studied in great detail. It is not clear whether there are cases in which it has great advantages.

10.7 Derivation of a Collective Hamiltonian

10.7.1 General Considerations

As we discussed in Section 10.2, the direct solution of the HW equation (10.5) can be quite cumbersome. In particular, the construction of the collective Hamiltonian (10.17) in analytical form is hardly possible in realistic cases. On the other hand, there are, at least for heavy nuclei, a number of suggestive approximations more or less based on the idea of transforming Eq. (10.5) into a differential equation, which looks similar to a Schrödinger equation in a collective variable q [FC 72, HW 72, BB 73, Ho 73, GG 75, Re 76a+b]. These theories have also been used to derive a collective Hamiltonian (as discussed in Chap. 1) from a microscopic point of view.

We start with a set of time-reversal invariant generating functions $|q\rangle = |\Phi(q)\rangle$, where q is a collective parameter; it characterizes, for instance, a certain deformation. The methods discussed in the following are based on two facts:

(i) In many cases, the overlap function $\mathcal{N}(q, q') = \langle q|q'\rangle$ is very small unless the two parameters q and q' are very close to each other, that is, $\mathcal{N}$ is a *rapidly decreasing* function with increasing distance $|q - q'|$. For instance, this is the case for the Lipkin model, where the function $\mathcal{N}$ in Eq. (10.32) depends only on $\varphi - \varphi'$ and is sharply peaked at the origin for large values of Ω (large particle number). The same is true for large particle numbers in the general case of two Slater determinants which differ from one another in a collective parameter, for example, in the deformation [HW 74]. This case will be discussed in more detail later (Sec. 10.7.4), and it will be shown that in many cases $\mathcal{N}$ has the form of a Gaussian in the variable $q - q'$. Only the width of this Gaussian may still depend on the center of mass $\frac{1}{2}(q + q')$.

(ii) We assume that the overlap functions $\mathcal{N}$ and $\mathcal{H}$ are *well behaved*—that is, differentiable—many times. This seems reasonable for many physical cases, and in particular when we have to deal with collective phenomena. It is not justified, however, when level crossings occur. There the overlap functions of HF Slater determinants make sudden jumps if there is a change in the occupation numbers [Re 77]. The occupation numbers v^2 in HFB functions are smeared out; this gives smoother but perhaps rapid changes of the functions $\mathcal{N}$ and $\mathcal{H}$. However, we must realize that in the HFB case we have to readjust the chemical potential λ; this is a nonlinear problem and causes additional difficulties. There may be more than one solution and sudden jumps between them. We shall therefore omit problems of level crossings in this section (see also Sec. 12.3.7).

There are essentially three different methods in order of increasingly simplifying assumptions: the *symmetric moment expansion* (SME), the *local approximation* (LA), and the *Gaussian overlap approximation* (GOA). We shall discuss them successively in this section. For the GOA we will present an independent derivation. Readers who are not interested in the details may therefore skip the next two subsections.

10.7.2 The Symmetric Moment Expansion (SME)

The two assumptions of differentiability and sharply peaked overlap functions can be used to transform the integral equation (10.5) into a second-order differential equation. In principle we have already applied similar techniques in Section 10.4 in the derivation of boson expansions. However, at that time it was a rather formal method and we ended up with a second-order differential equation in many hundreds of variables z_{ik}. We shall now restrict ourselves to one (or a few) collective variables q.

Using the identity

$$|q'\rangle =: e^{-i(q'-q)P/\hbar}: |q\rangle \tag{10.91}$$

with*

$$P = -\frac{\hbar}{i}\frac{\partial}{\partial q} \tag{10.92}$$

and the dots meaning normal ordering, so that powers of $(q'-q)$ are always to the left side of P, we can write for the integral equation (10.5) (we drop the normal ordering points in the following):

$$\int dq'(\mathcal{H}(q,q') - E\mathcal{N}(q,q'))e^{-i(q'-q)P/\hbar}f(q) = 0. \tag{10.93}$$

As we assume that the norm overlap is a rapidly decreasing function of $|q'-q|$, we see by inspection of the Onishi formula (10.39) that the Hamilton overlap $\mathcal{H}(q,q')$ should also be quite a rapidly decreasing function of $|q-q'|$. We are therefore allowed to develop the exponentials up to second order. We first introduce the variable

$$s = q - q' \tag{10.94}$$

and gain from (10.5)

$$\int_{-\infty}^{+\infty} ds \langle q|H-E|q-s\rangle e^{isP/\hbar}f(q) = 0. \tag{10.95}$$

As the second-order differential operator we will obtain on the left-hand side of Eq. (10.95) is not, term by term, a symmetric operator [BB 73], it is preferable to symmetrize this equation first [Ho 73]:

$$\int_{-\infty}^{+\infty} ds\, e^{isP/2\hbar}\langle q+s/2|H-E|q-s/2\rangle e^{isP/2\hbar}f(q) = 0. \tag{10.96}$$

The (still) exact equation (10.96) is now approximated by expanding the exponentials up to second order in s:

$$(H_0 - \tfrac{1}{2}\overline{H_2P^2})f = E(N_0 - \tfrac{1}{2}\overline{N_2P^2})f, \tag{10.97}$$

with the moments

$$\left.\begin{matrix} H_n(q) \\ N_n(q) \end{matrix}\right\} = \frac{1}{\hbar^n}\int_{-\infty}^{+\infty} ds\, s^n \langle q+s/2|\begin{matrix} H \\ 1 \end{matrix}|q-s/2\rangle \tag{10.98}$$

and the definition of a symmetric ordered product of operators [Re 76b], viz.

$$\begin{aligned}\overline{A(q)P^2} &:= \frac{1}{4}(AP^2+2PAP+P^2A) = \frac{1}{4}[P,[A,P]_+]_+ \\ &= \sum_{n=0}^{\mu}\frac{1}{2^\mu}\binom{\mu}{n}P^nAP^{\mu-n}\Big|_{\mu=2}.\end{aligned} \tag{10.99}$$

The odd moments vanish identically, $H_1 = N_1 = 0$, since we have assumed that the set of generating states $|\Phi(q)\rangle$ is invariant under time reversal, which means that the overlap functions $\mathcal{N}$ and $\mathcal{H}$ are real. The only quantities we have to know are the zeroth and second moments of the norm and the Hamiltonian overlap (10.98).

*The operator P acts on the parameter q and not on the particle coordinates (x) in the wave function $|\Phi(q)\rangle$. It is therefore convenient to use a minus sign in its definition. In the case of pure translation, where q is the origin of the new coordinate system, we get $|\Phi(\mathbf{x}_i, \mathbf{q})\rangle = |\Phi(\mathbf{x}_i - \mathbf{q}, 0)\rangle$ and $\sum(\hbar/i)(\partial/\partial\mathbf{x}_i)|\Phi\rangle = -(\hbar/i)(\partial/\partial\mathbf{q})|\Phi\rangle$. In the present context the minus sign does not matter, since the Hamiltonian we are going to derive is quadratic in P.

Equation (10.97) is a Hermitian second-order differential equation for $f(q)$. However, it is still not of the usual Schrödinger type. To arrive at such an equation we have to decompose the operator $(N_0 - \frac{1}{2}\overline{N_2 P^2})$ in the following way.

$$\left(N_0 - \tfrac{1}{2}\overline{N_2 P^2}\right) = (K^{1/2})^{+}(K^{1/2}). \tag{10.100}$$

Starting from the ansatz

$$K^{-1/2} = N_0^{-1/2}\left(1 + \overline{K_2(q)P^2} + \overline{K_4(q)P^2} + \cdots\right), \tag{10.101}$$

we can insert into Eq. (10.100) and bring the result in each order to the symmetric form defined by Eq. (10.99). This allows us to determine the functions K_n. We will see that in most cases we only need them up to $n \leqslant 4$. In practice, the calculations are quite lengthy. Also, we usually neglect derivatives $(\partial^n/\partial q^n)N_0$ and $(\partial^n/\partial q^n)N_2$ for $n \geqslant 2$ and stop after second order. This is consistent with the quadratic approximation (10.97).

For the sake of simplicity, in the following we will neglect all derivatives of N_0 and N_2. This is realized in all the cases where (q, q') depends only on the difference $(q - q')$. In this case, up to order P^2 we get:

$$K^{-1/2} = N_0^{-1/2}\left(1 + \frac{1}{4}\frac{N_2}{N_0}P^2\right). \tag{10.102}$$

In the next step we have to calculate the collective Hamiltonian $K^{-1/2+}(H_0 - \frac{1}{2}\overline{H_2 P^2})K^{-1/2}$. Again, we have to bring all orders of P^2 into the symmetric form (10.99).* We usually neglect terms of order higher than P^2 and the derivatives $(\partial^n/\partial q^n)H$ for $n \geqslant 2$. At this point, this is certainly not well justified. We should calculate the corresponding higher-order coefficients and see if they are small. In fact, we can show, on the basis of the Lipkin model, that they are of the order $(1/\Omega)^2$ [Ho 73], but this is probably not very significant.

In general, we expect that this method is justified for motions with large amplitudes, but not very large velocities.

Under these conditions we get from Eq. (10.102) the following Schrödinger equation for the function $g(q) = K^{1/2}f(q)$.

$$\left(\overline{\frac{1}{2M(q)}P^2} + V(q)\right)g(q) = Eg(q), \tag{10.103}$$

with

$$V(q) = \frac{H_0}{N_0} - \frac{\hbar^2}{8}\frac{N_2}{N_0}\cdot\frac{H_0''}{N_0} \qquad \text{and} \qquad \frac{1}{M(q)} = \frac{H_0}{N_0}\frac{N_2}{N_0} - \frac{H_2}{N_0}, \tag{10.104}$$

where the primes mean differentiation with respect to q. We have also neglected in $1/M$ the derivatives H_0'', H_2'' consistent with the fact that P^2 is already of second order in the momentum.

Equation (10.103), together with Eq. (10.104) is the consistent transformation of Eq. (10.5) into a Hermitian second-order differential equation. This is a full quantum mechanical equation. Since the mass parameter M depends on the coordinate q, the relative order of P and M in Eq. (10.103) is important. Here we

*These calculations are rather simple if we use the formula [Re 76b] [for any function $X(q)$, X'' means $(\partial^2/\partial q^2)X$]:

$$P^l\overline{XP^m}P^n + P^n\overline{XP^m}P^l = 2\overline{XP^{l+m+n}} + \tfrac{1}{4}((n+l)-(n-l)^2)\hbar^2\overline{X''P^{l+m+n-2}}.$$

have used the prescription (10.102), which is a result of the symmetric expansion (10.97). We should be aware, however, that this is not the only possibility, and to require the same ordering after the inversion (10.101) seems somewhat arbitrary. Flocard and Vautherin [FV 74], for example, give a slightly different formulation.

In the formulae (10.104) we neglected the coordinate dependence of N_0 and N_2. This can be included [Re 76b] yielding rather lengthy expressions of the same structure.

In Eq. (10.103) the zeroth and second moments of the norm and Hamilton overlap are unknown. Since these functions must be known over their whole range of nonlocality, we call this the *global approximation*. It can be checked immediately that our harmonic oscillator model (Sec. 10.5) is exactly reproduced by Eq. (10.103). This is only true, however, in lowest order. Including higher orders in the expansion (10.97) adds small corrections. In fact, we have to go up to infinite order in the moment expansion (10.97) to get the proper weight function f of Eq. (10.56). Because of its global nature, for more realistic problems Eq. (10.103) is still quite complicated. Therefore, we try to reduce its complexity further by introducing the local approximation.

10.7.3 The Local Approximation (LA)

In Section 10.7.1 we put forward arguments that the Hamilton overlap should not be completely different in its q-dependence from the norm overlap (see also (10.32)). It should, for instance, also be a rapidly decreasing function if this is the case for the norm overlap. Therefore, it seems reasonable to stop the following (in principle, exact expansion) after second-order terms:

$$\mathcal{H}\left(q+\frac{s}{2}, q-\frac{s}{2}\right)=\left\{A(q)+B(q)\frac{\partial^2}{\partial s^2}+\cdots\right\}\mathcal{N}\left(q+\frac{s}{2}, q-\frac{s}{2}\right). \quad (10.105)$$

Because of time reversal invariance, there are no odd derivatives in Eq. (10.105). Expanding both sides of this equation in powers of s yields a system of equations for the coefficients $A, B, \ldots$. To second order, we obtain the following expressions.

$$A(q)=\left.\frac{\mathcal{H}\cdot\mathcal{N}^{iv}-\mathcal{H}''\cdot\mathcal{N}''}{\mathcal{N}\cdot\mathcal{N}^{iv}-\mathcal{N}''\cdot\mathcal{N}''}\right|_{s=0}, \quad (10.106)$$

$$B(q)=-\left.\frac{\mathcal{H}\cdot\mathcal{N}''-\mathcal{H}''\cdot\mathcal{N}}{\mathcal{N}\cdot\mathcal{N}^{iv}-\mathcal{N}''\cdot\mathcal{N}''}\right|_{s=0}. \quad (10.107)$$

The upper primes mean differentiation with respect to s. Using $\mathcal{N}(q, q)=1$ and inserting the expression (10.105) with Eqs. (10.106) and (10.107) into the integrals (10.98), we obtain for the first two moments of the Hamiltonian ($\int ds\, \mathcal{N}''=0$):

$$H_0(q)=N_0\cdot A, \quad (10.108)$$

$$H_2(q)=\frac{N_2}{N_0}\cdot H_0+\frac{2}{\hbar^2}N_0\cdot B. \quad (10.109)$$

Using the approximate expressions (10.108) and (10.109) for H_0 and H_2, we have achieved a reduction of the information we need about the Hamilton overlap to that of a knowledge of its local $\mathcal{H}(q, q)$ and "slightly nonlocal" values $\mathcal{H}''|_{s=0}$. This is naturally an advantage over the expression (10.98) where we had to know the Hamilton overlap over its entire range of nonlocality. Only for the norm overlap do we still have to know its nonlocality over the entire range, since its zeroth and

second moments enter Eq. (10.104). The theory developed so far [Re 76a+b] is equivalent to that of Giraud and Grammaticos [GG 74a, GG 75]. It still gives the exact result for the harmonic oscillator model (Sec. 10.5). However, it is a global approximation of the norm overlap. It may be that for specific cases where the norm overlap is easier to calculate than the Hamilton overlap, the local approximation is useful; however, in general, this is also a very difficult task. In the next section we will therefore develop an approximation scheme for the norm overlap.

10.7.4 The Gaussian Overlap Approximation (GOA)

In order to derive simple expressions for the parameters of the collective Hamiltonian, the Gaussian overlap approximation is widely used [HW 53, JS 64, Ka 68, Vi 75]: We assume that the norm overlap $\mathcal{N}(a, a')$ is in the center of mass coordinates

$$q = \frac{a+a'}{2}, \qquad s = (a-a') \tag{10.110}$$

of the form

$$\mathcal{N}(a,a') = \mathcal{N}\left(q+\frac{s}{2}, q-\frac{s}{2}\right) = e^{-\frac{1}{2}\gamma(q)s^2}. \tag{10.111}$$

With the definition (10.92) for the operator $P = -(\hbar/i)(\partial/\partial q)$ we find ($|q\rangle \equiv |\Phi(q)\rangle$),

$$\hbar^2\gamma(q) = \langle q|P^2|q\rangle. \tag{10.112}$$

The expectation value of P vanishes because of time reversal invariance

$$\langle q|P|q\rangle = 0. \tag{10.113}$$

Before we apply the Gaussian approximation, we first have to discuss the conditions under which it is valid. To be more precise, we suppose for a moment that $|q\rangle$ is a set of Slater determinants depending on a collective parameter q, as may be obtained, for instance, by a constrained Hartree–Fock calculation (see Section 7.6).

Thouless' theorem tells us that we can represent a function in the vicinity of $|q\rangle$ as [(E.40) and (E.26)]

$$|q+\tfrac{s}{2}\rangle = e^{-is\hat{P}/2\hbar}|q\rangle = \mathcal{N}\left(q, q+\frac{s}{2}\right)\cdot e^{s\hat{Z}/2\hbar}|q\rangle, \tag{10.114}$$

where $\hat{P}$ is a Hermitian operator with only *ph* and *hp* matrix elements. It is obviously a representation of the operator $-(\hbar/i)(\partial/\partial q)$ in the Hilbert space. The operator $\hat{Z}$ has only *ph* matrix elements. As long as s is small, we can assume $\hat{P}$ and $\hat{Z}$ to be independent of s and find, from Eq. (E.48), that $\hat{Z}$ is just the *ph* part of $\hat{P}$. Both operators depend on q.

Using the formula of Onishi (E.51), we find, up to a phase,

$$\mathcal{N}\left(q+\frac{s}{2}, q-\frac{s}{2}\right) = \exp\left\{\mathrm{Tr}\left(-\mathrm{Artanh}\frac{s^2}{2}A\right)\right\}, \tag{10.115}$$

where the matrix A is given by

$$A_{ii'} = \sum_m \frac{Z^*_{im} Z_{mi'}}{\hbar^2} . \tag{10.116}$$

For small values of s, we can expand the exponent of Eq. (10.115) and obtain the Gaussian overlap* (10.111)

$$\mathcal{N}\left(q+\frac{s}{2}, q-\frac{s}{2}\right) = \exp\left(-\frac{1}{2}\gamma(q)s^2\right), \tag{10.117}$$

with

$$\gamma(q) = \mathrm{Tr}\, A = \sum_{mi} |P_{mi}|^2/\hbar^2 = \frac{1}{\hbar^2}(\langle q|\hat{P}^2|q\rangle - \langle q|\hat{P}|q\rangle^2). \tag{10.118}$$

We see that the crucial parameter in this approximation is the fluctuation of the operator $\hat{P}$ in the Slater determinant $|q\rangle$. If it is large compared to $\hbar/s$, we can neglect higher terms in Eq. (10.117) and end up with the Gaussian shape. For a general collective operator, $\hat{P}$, γ grows roughly with the particle number. Therefore, the Gaussian overlap approach is justified for heavy systems.

The function $\gamma(q)$ determines some sort of a metric along the path $|q\rangle$. It turns out to be convenient to introduce a scale transformation for an absolute coordinate α in such a way that the norm has a constant width γ_0. This is achieved by the transformation

$$\gamma_0 (d\alpha)^2 = \gamma(a)(da)^2, \qquad \alpha = \int^a \sqrt{\gamma(a')/\gamma_0}\, da'. \tag{10.119}$$

In first order we then get, in the exponent of Eq. (10.117),

$$\gamma\left(\frac{a+a'}{2}\right)(a-a')^2 = \gamma_0(\alpha-\alpha')^2.$$

In the following we assume that we are working with this absolute coordinate.

In the case of a Gaussian overlap, it is rather simple to derive an equation of motion in the coordinate q [Vi 75, GG 75, Ka 73, OU 75], because we can now calculate $\mathcal{N}^{1/2}$ in Eq. (10.10) analytically. It is easy to show that $\mathcal{N}(a, a')$ can be written as

$$\mathcal{N}(a, a') = \int dq\, \sqrt{\gamma_0}\, \mathcal{N}^{1/2}(a, q)\mathcal{N}^{1/2}(q, a'),$$

with

$$\mathcal{N}^{1/2}(a, q) = \left(\frac{2}{\pi}\right)^{1/4} \exp\left(-\gamma_0 \cdot (a-q)^2\right). \tag{10.120}$$

The expectation values of the Hamiltonian H for the GCM function

$$|\Psi\rangle = \int da\, f(a)|a\rangle \tag{10.121}$$

* In the case where $|q\rangle$ also has time odd components, we gain an additional phase [see Eq. (11.79)].

are then given by:

$$\langle\Psi|H|\Psi\rangle = \int da\, da'\, f^*(a)\mathcal{N}(a,a')h(a,a')f(a')$$

$$= \int da\, da'\, f^*(a)\int dq\, \sqrt{\gamma_0}\, \mathcal{N}^{1/2}(a,q)h(a,a')\mathcal{N}^{1/2}(q,a')f(a'). \quad (10.122)$$

In the next step, we replace $h(a,a')$ by a differential operator, the collective Hamiltonian

$$h(a,a') \triangleq \mathcal{H}_{\text{coll}}\left(q, \frac{\partial}{\partial q}\right), \quad (10.123)$$

which no longer depends on a and a'. To achieve this, we use the fact that $\mathcal{N}^{1/2}(a,q)$ is sharply peaked at $a=q$ and expand $h(a,a')$ at the point $a=a'=q$ up to second order in the differences $(a-q)$ and $(a'-q)$. Here we again use the property that $h(a,a')$ is a well-behaved smooth function:

$$h(a,a') = h + h_x\cdot(a-q) + h_y\cdot(a'-q) + \tfrac{1}{2}\big(h_{xx}\cdot(a-q)^2 + 2h_{xy}\cdot(a-q)(a'-q) + h_{yy}\cdot(a'-q)^2\big), \quad (10.124)$$

where

$$h = h(q,q) = \langle q|H|q\rangle, \quad h_x = \frac{\partial}{\partial a}h(a,a')\Big|_{a=a'=q}, \quad h_y = \frac{\partial}{\partial a'}h(a,a')\Big|_{a=a'=q}$$

$$h_{xx} = \frac{\partial^2}{\partial a^2}h(a,a')\Big|_{a=a'=q}; \quad h_{xy} = \frac{\partial^2}{\partial a\,\partial a'}h(a,a')\Big|_{a=a'=q}; \quad h_{yy} = \cdots. \quad (10.125)$$

The linear derivations do not contribute to the final result because of time-reversal invariance. To see this and to get a simple expression for the second derivatives, we write the full Hamilton overlap as [see Eq. (10.91)],

$$\mathcal{H}(a,a') = \langle q|e^{i(a-q)\hat{P}/\hbar}He^{-i(a'-q)\hat{P}/\hbar}|q\rangle, \quad (10.126)$$

and get [with a similar expression for $\mathcal{N}(a,a')$] ($\mathcal{N}(a,a)=1$)

$$\frac{\partial}{\partial a}h(a,a')_{a=a'=q} = \frac{\partial}{\partial a}\left(\frac{\mathcal{H}(a,a')}{\mathcal{N}(a,a')}\right)_{a=a'=q}$$

$$= \frac{i}{\hbar}\langle q|\hat{P}H|q\rangle = 0 \quad (10.127)$$

and

$$h_x - h_y = \frac{i}{\hbar}\langle q|\hat{P}H + H\hat{P}|q\rangle = 0.$$

In the same way, we find

$$h_{xx} = \frac{1}{\hbar^2}\langle q|\hat{P}^2H|q\rangle_L := -\frac{1}{\hbar^2}(\langle q|\hat{P}^2H|q\rangle - \langle q|H|q\rangle\langle q|\hat{P}^2|q\rangle), \quad h_{xx} = h_{yy} \quad (10.128)$$

and so on. The index L (linked) in this context* means that we have to subtract the product of the corresponding matrix elements for the norm and the expectation value of H. Similar expressions can be derived for h_{xy} and h_{yy}. In particular, we get

$$\frac{1}{4}(h_{xx}-2h_{xy}+h_{yy}) := \frac{-1}{\hbar^2}\langle q|\hat{P}^2 H|q\rangle_L = \frac{-1}{4\hbar^2}\langle q|[\hat{P},[H,\hat{P}]_+]_+|q\rangle_L. \tag{10.129}$$

We now come back to Eq. (10.122) and insert (10.124). Since the functions $\mathfrak{N}^{1/2}(a,q)$ have Gaussian form (10.120), we can express terms $(a-q)\cdots$ by

$$(a-q)\mathfrak{N}^{1/2} = \frac{1}{2\gamma_0}\frac{\partial}{\partial q}\mathfrak{N}^{1/2}; \qquad (a-q)^2\mathfrak{N}^{1/2} = \left(\frac{1}{4\gamma_0^2}\frac{\partial^2}{\partial q^2}+\frac{1}{2\gamma_0}\right)\mathfrak{N}^{1/2}. \tag{10.130}$$

After partial integrations we return to the general coordinates (10.119) and obtain, if we neglect higher than second-order derivatives of h (we have to realize that despite of (10.127) the term $\partial/\partial q\, h_x = h_{xx}+h_{xy}$ contributes):

$$\langle\Psi|H|\Psi\rangle = \int \sqrt{\gamma}\, dq\; g^*(q)\mathcal{H}_{\text{coll}}\left(q,\frac{\partial}{\partial q}\right)g(q), \tag{10.131}$$

where

$$g(q) = \int \mathfrak{N}^{1/2}(q,a)f(a)\, da. \tag{10.132}$$

The collective Hamiltonian is given by

$$\mathcal{H}_{\text{coll}} = -\frac{1}{\sqrt{\gamma}}\frac{\partial}{\partial q}\sqrt{\gamma}\,\frac{\hbar^2}{2M(q)}\frac{\partial}{\partial q}+V(q) \tag{10.133}$$

with the inertial parameter

$$\frac{1}{M(q)} = \frac{-1}{2\gamma^2}(h_{xx}-h_{xy}) = \frac{\langle q|[\hat{P},[H,\hat{P}]_+]_+|q\rangle_L}{4\langle q|\hat{P}^2|q\rangle^2} \tag{10.134}$$

and the potential energy

$$V(q) = \tilde{V}(q)-\epsilon_0(q), \qquad \tilde{V}(q) = \langle q|H|q\rangle, \tag{10.135}$$

$$\epsilon_0(q) = \frac{h_{xy}}{2\gamma} = \frac{\langle q|\hat{P}^2|q\rangle}{2M(q)}+\frac{\hbar^2}{8\langle q|\hat{P}^2|q\rangle}\frac{\partial^2}{\partial q^2}\tilde{V}(q). \tag{10.136}$$

$$\hbar^2\frac{\partial^2}{\partial q^2}\tilde{V}(q) = -\langle q|[\hat{P},[\hat{P},H]_-]_-|q\rangle.$$

We end up with a collective Hamiltonian in the variable q. Its potential energy $V(q)$ is not simply the static expectation value of the Hamiltonian, but is corrected for the value $\epsilon_0(q)$, which is called the zero-point energy [Re 75c, 78]; because an oscillating wave packet $|q\rangle$ of the form (10.84) in

* For the general definition of linked matrix elements see, for instance, [VS 71].

the vicinity of the minimum of $V(q)$ would have just this energy (in this case $\hbar^2/\langle q|\hat{P}^2|q\rangle = 4\langle q|\hat{Q}^2|q\rangle$). Equations (10.133)–(10.136) are again exact for our harmonic oscillator model of Section 10.5. They also turn out to be strictly equivalent to Eqs. (10.103)–(10.109), if we specialize to Gaussian norm.*

If we solve the equation of motion with (10.133) for an oscillation in a potential $\tilde{V}(q)$, we get a ground state having a zero-point energy higher than the HF minimum in the potential V. Calculations [Go 77] showed that the potential $V = \tilde{V} - \epsilon_0$ has a ground state which lies just below the HF value (see Fig. 12.6).

The inertia parameter M in Eq. (10.134) turns out to be coordinate dependent. It is usually called *Yoccoz-inertia* for reasons which we will see in Section 11.4. In the case of pure translations, it is not the bare mass of the nucleus. It is a great theoretical disadvantage of the primitive version of the GCM theory that it does not yield the proper mass. This does not, however, mean that the results of this theory are necessarily bad.

Several attempts have been made to improve the theory in order to obtain the proper mass. Peierls and Thouless [PT 62] proposed to use a GCM with two generator coordinates, a coordinate q and a momentum p. As discussed in Section 11.4, this procedure yields Galilei invariant wave functions in the translational case and the proper mass.

On the other hand, it is quite cumbersome to use two generator coordinates.† Holzwarth and Villars [Vi 75] therefore proposed to generalize the GCM ansatz in the following way.

$$|\Psi\rangle = \int dq\, e^{\hat{S}(q)\frac{\partial}{\partial q}} |\Phi(q)\rangle f(q),$$

where $\partial/\partial q$ acts only on the weight function $f(q)$, whereas the single-particle operator $\hat{S}(q)$ acts on the wave function. With an appropriate choice for the operator $\hat{S}$, we obtain in this way the proper mass parameter using only one generator coordinate.

10.7.5 The Lipkin Model

In order to give a demonstrative example of this theory, we again take the Lipkin model (Sec. 10.3). For large particle numbers Ω the norm overlap function (10.32) behaves like a Gaussian. With $\phi = \varphi - \varphi'$, $\Psi = \frac{1}{2}(\varphi + \varphi')$; we get

$$\mathcal{N}(\phi) = \exp\left(-\frac{1}{8}\Omega\phi^2\right) + O\left(\frac{1}{\Omega}\right) \tag{10.137}$$

*In order to verify this, we have to realize that $(\hbar^2/2\langle\hat{P}^2\rangle)\partial^2/\partial q^2\mathcal{H}''|_{s=0} \simeq -(1/2)\,\partial^2/\partial q^2\mathcal{H}|_{s=0}$.

†In this context, see also [HW 72, 74, GG 79].

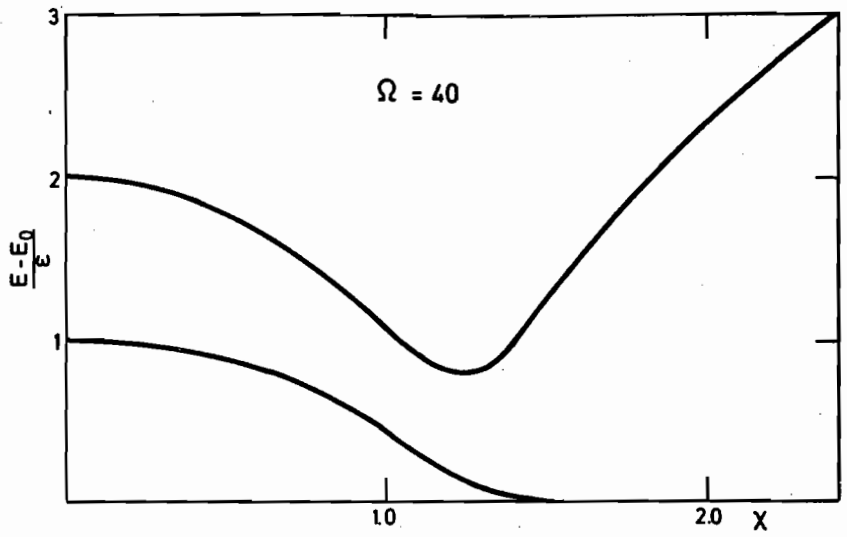


Figure 10.1. Excitation energies of the first and second excited states for $\Omega = 40$. (From [Ho 73].)

and

$$\mathcal{K}(\Psi + \tfrac{1}{2}\phi, \Psi - \tfrac{1}{2}\phi) = \left\{ \mathcal{K}(\Psi, \Psi) + \frac{\Omega}{4} B(\Psi) + B(\Psi) \frac{\partial^2}{\partial \phi^2} \right\} \mathcal{N}(\phi); \quad (10.138)$$

with

$$B(\Psi) = -\frac{\epsilon}{\Omega}\left(\cos\Psi + \chi(1 + \sin^2\Psi)\right). \quad (10.139)$$

From Eqs. (10.104) and (10.108) or (10.134) and (10.135), we get for the mass

$$\frac{1}{M(\Psi)} = \frac{2\epsilon}{\Omega}\left(\cos\Psi + \chi(1 + \sin^2\Psi)\right) \quad (10.140)$$

and for the potential

$$V(\Psi) = -\frac{\epsilon}{2}\Omega\left(\cos\Psi + \frac{1}{2}\chi\sin^2\Psi\right) - \frac{\epsilon}{4}(2\cos\Psi + 3\chi\sin^2\Psi) + O\left(\frac{1}{\Omega}\right). \quad (10.141)$$

These two quantities are shown as a function of $\alpha = \Psi/2$ in Fig. 5.2. The dashed lines give the potential V and the chain curves the inverse mass parameter $1/M$.

The solution of the collective equation for the two first excited states with these potentials are given in Fig. 10.1 and compared with the exact ones.

The approximate solutions and the exact ones cannot be distinguished. This is not very astonishing, since we know that the GCM is able to produce the exact result (Sec. 10.4) and the terms neglected in the symmet-

ric momentum expansion and the Gaussian overlap approach in this model are of the order $1/\Omega$.

10.7.6 The Multidimensional Case

For the sake of simplicity we have so far treated only one generator coordinate. In the general case of several coordinates $\mathbf{a}=(a_1,\ldots,a_f)$, Kamlah [Ka 73] and Onishi et al. [OU 75] have used a general Gaussian ansatz for the norm overlap

$$\mathcal{N}(\mathbf{a},\mathbf{a}')=\mathcal{N}\left(\mathbf{q}+\frac{\mathbf{s}}{2},\mathbf{q}-\frac{\mathbf{s}}{2}\right)=\exp\left(-\frac{1}{2}s^2(\mathbf{q},\mathbf{s})\right), \tag{10.142}$$

where $\mathbf{q}=\frac{1}{2}(\mathbf{a}+\mathbf{a}')$, $\mathbf{s}=\mathbf{a}-\mathbf{a}'$, and

$$s^2=\gamma_{ij}(\mathbf{q})\cdot s^i s^j \tag{10.143}$$

is the squared geodesic distance in a curved space characterized by the coordinates $\mathbf{a}$. Applying the theory of Riemann spaces one can proceed as in Section 10.7.4 and end up with a collective Hamiltonian

$$\mathcal{H}_{\text{coll}}=-\frac{1}{\sqrt{\gamma}}\frac{\partial}{\partial q^i}\sqrt{\gamma}\left(\frac{\hbar}{2M}\right)^{ij}\frac{\partial}{\partial q^j}+\tilde{V}(\mathbf{q}), \tag{10.144}$$

where $\gamma(\mathbf{q})$ is the determinant of the metric tensor γ_{ik} defined by the ansatz (10.143). The inertia tensor and the potential have a form analogous to Eqs. (10.134 ff):

$$\left(\frac{\hbar^2}{2M(\mathbf{q})}\right)_{ij}=-\frac{1}{4}\left(\frac{\Delta}{\Delta q^i}\frac{\Delta}{\Delta q^j}h(\mathbf{q},\mathbf{q}')-\frac{\Delta}{\Delta q^i}\frac{\Delta}{\Delta q'^j}h(\mathbf{q},\mathbf{q}')\right)_{\mathbf{q}=\mathbf{q}'}, \tag{10.145}$$

$$V(\mathbf{q})=\tilde{V}(\mathbf{q})-\epsilon_0(\mathbf{q});\qquad \tilde{V}(\mathbf{q})=\langle\mathbf{q}|H|\mathbf{q}\rangle,$$

$$\epsilon_0(\mathbf{q})=\gamma_{ij}\left(\frac{\hbar^2}{2M}\right)^{ij}+\frac{1}{8}\gamma^{ij}\frac{\Delta}{\Delta q^i}\frac{\Delta}{\Delta q^j}V(\mathbf{q}),$$

where $\Delta/\Delta q^i$ are covariant derivations of the tensor fields as defined, for instance, in [LL 59, vol. II, Chap. 10]. The operator of the kinetic energy is similar to the operator we get from Pauli quantization [Eq. (1.53)]. In the latter case, however, the inertia tensor is identical with the metric tensor. Now we have to distinguish between the *metric* determined completely by the kinematical properties, that is, by $\mathcal{N}$, and the *inertia* given by the dynamical behavior of $\mathcal{H}$.

10.8 The Choice of the Collective Coordinate

In Section 10.4 we saw that the ansatz (10.2) contains, in principle, the exact answer to the many-body problem if we choose enough generator coordinates. Unfortunately, this is by far too large a problem to be handled exactly, and in practice we are restricted to the treatment of one or perhaps a few generator coordinates at a time. It is therefore very important to choose the proper generator coordinate, that is, the most suitable set $|q\rangle=|\Phi(q)\rangle$. In the following, we again restrict ourselves to a real coordinate q with time even functions $|q\rangle$.

The set of generating functions $|\Phi(q)\rangle$ represents a path in the multidimensional energy surface

$$E[\Phi]=\langle\Phi|H|\Phi\rangle.$$

To determine this path, we again apply the variational principle and require that the expectation value of the energy in the state

$$|\Psi\rangle=\int f(q)|\Phi(q)\rangle\,dq \tag{10.146}$$

is stationary not only with respect to variations of the weight function f, but also with respect to the path $|\Phi(q)\rangle$. This *double variational method* gives us a set of coupled equations [HY 74, Vi 75]

$$\int dq'\,\langle\Phi(q)|H-E|\Phi(q')\rangle f(q')=0, \tag{10.147a}$$

$$\int dq\,dq'\,f^*(q)\langle\delta\Phi(q)|H-E|\Phi(q')\rangle f(q')=0, \tag{10.147b}$$

which, in principle, have to be solved by iteration.

At a point q, the variation $|\delta\Phi\rangle$ is not completely arbitrary. We can only allow for changes $|\delta\Phi\rangle_\perp$ that are orthogonal to the path itself. A variation parallel to the path could be absorbed in a variation of the weight function.

Since the length of $|\delta\Phi\rangle_\perp$ is arbitrary, Eq. (10.147b) may be written as

$$\int dq'\,{}_\perp\langle\delta\Phi(q)|H-E|\Phi(q')\rangle f(q')=0. \tag{10.148}$$

In the next step, we make two essential assumptions:

(i) We again use the fact that the overlap functions $\langle\Phi(q)|\ldots|\Phi(q')\rangle$ are sharply peaked at $q=q'$, such that for fixed q only a certain q' region contributes to the integral (10.148).

(ii) We assume that in this q'-region, the path does not bend too strongly, such that $|\delta\Phi(q)\rangle$ is orthogonal to the path in this whole region.

Under these conditions, the second equation (10.147b) decouples from the first one (10.147a), and for variations $|\delta\Phi\rangle_\perp$ that do not change the norm of $|\Phi\rangle$ [Vi 75] we find

$${}_\perp\langle\delta\Phi|H|\Phi\rangle=0. \tag{10.149}$$

The variation $|\delta\Phi\rangle_\perp$ is restricted to be orthogonal to the path. A variation parallel to the path is given by

$$|\delta\Phi\rangle_\parallel=|\Phi(q+\delta q)\rangle-|\Phi(q)\rangle=-\delta q\frac{i}{\hbar}\hat{P}|\Phi(q)\rangle. \tag{10.150}$$

The operator $\hat{P}$ is a representation of $-(\hbar/i)\partial/\partial q$ in the many-body Hilbert space. It acts on the particle coordinates in $|\Phi\rangle$ but not on the parameter q. Variations orthogonal to the path now have the form

$$|\delta\Phi\rangle_\perp=|\delta\Phi\rangle-\hat{P}|\Phi\rangle\frac{\langle\Phi|\hat{P}|\delta\Phi\rangle}{\langle\Phi|\hat{P}^2|\Phi\rangle}, \tag{10.151}$$

where $|\delta\Phi\rangle$ is an unrestricted variation. Variations of the potential energy $V(q)=\langle\Phi(q)|H|\Phi(q)\rangle$ parallel to the path are given by

$$\frac{dV}{dq}=+\frac{i}{\hbar}\langle\Phi|[\hat{P},H]|\Phi\rangle \tag{10.152}$$

and those orthogonal to it have to vanish:

$$_{\perp}\langle\delta\Phi|H|\Phi\rangle=\langle\delta\Phi|H-\frac{\hbar}{2i}\frac{dV}{dq}\frac{\hat{P}}{\langle\Phi|\hat{P}^2|\Phi\rangle}|\Phi\rangle=0. \tag{10.153}$$

In deriving this equation, we have used (10.152) and the fact that $|\Phi\rangle$ is time even. The problem (10.153) is a constrained variational principle, but the constraining operator is not Hermitian; we therefore restrict ourselves to cases in which $\hat{P}$ can be decomposed.

$$\hat{P}=i(A^+-A) \qquad \text{with } A|\Phi\rangle=0. \tag{10.154}$$

Examples are Slater determinants $|\Phi(q)\rangle$, where we have $A^+=(1/i)\sum_{mi}P_{mi}a_m^+a_i$, or RPA wave functions, where A^+ is a creation operator for a boson (see also [ABC 77b]).

In such cases, we can define a Hermitian constraining operator

$$\hat{Q}=\frac{\hbar}{2\langle\Phi|\hat{P}^2|\Phi\rangle}(A^++A) \tag{10.155}$$

and from Eqs. (10.152) and (10.153) get

$$\langle\delta\Phi|H-\frac{dV}{dq}\cdot\hat{Q}|\Phi\rangle=0, \tag{10.156}$$

which gives us the optimal path. It is determined in such a way that the energy surface has a minimum with respect to all variations perpendicular to it. This means, for instance, that it follows the bottom of the fission valley (see Sec. 7.6). Equation (10.156) is a constrained variational problem. The constraining operator $\hat{Q}$ is not imposed on the system, but is determined self-consistently by the system itself. It does not push the system into unphysical regions, but acts like an inertial force in the sense of d'Alembert [Vi 77].

The actual calculation of $|\Phi(q)\rangle$ requires a double iteration: Starting with some initial value $\hat{Q}_0$ for the constraining operator, we have to determine $|\Phi_0(q)\rangle$ by a constrained HF calculation. In a second step, we have to calculate the operator $\hat{P}$ at each point q. It can be done, for instance, by a solution of Eq. (8.106) in linear response theory [see Eqs. (8.136) and (12.120)]. From the operator $\hat{P}$ a new constraint $\hat{Q}$ is determined. This has to be continued until convergence is achieved.

At the end, we have to solve the Hill–Wheeler equations (10.5) for the determined generating functions $|\Phi(q)\rangle$.

This double iteration, naturally, is extremely time consuming, and the prescription of a self-consistent constraint in the GCM has up to now been tested only on a three-level Lipkin model [HY 74] and in the $s-d$ shell [PL 77] with quite promising results. For heavier nuclei the gradient method described in Chapter 7 could facilitate the problem.

10.9 Application of the Generator Coordinate Method for Bound States

In the last few years, the GCM has been applied to many problems of nuclear structure, such as giant resonances and shape and pairing vibrations, and to coupling mechanisms between different modes. In most cases, generating functions with proper symmetries were used, such as Slater determinants projected onto good angular momentum or particle number (see Chap. 11). The Hill–Wheeler equation (10.5) is usually solved by discretization. In many cases, very few mesh points were chosen. Therefore, there were no problems with zero eigenvalues of the norm. We will discuss only a few examples in the following.

10.9.1 Giant Resonances

The breathing mode and quadrupole oscillations in *light nuclei* have been investigated using Slater determinants $\Phi_{JMP}(b, \delta)$ of deformed oscillator functions projected onto good angular momentum J, M and linear momentum $P=0$; they depend on the oscillator lengths $b=(b_x^2 b_z)^{1/3}$ and the deformation $\delta = b_x/b_z$ [CBA 73, AC 75]. For $\delta = 1$, the parameter b describes a scaling of the radial coordinate, and therefore a breathing mode. (It has already been used in a very early paper by Ferrell and Visscher [FV 56]). The coordinate δ describes the quadrupole oscillation. The Hill–Wheeler Eq. (10.5) is solved by discretization (up to 50 mesh points) with the interaction of Brink and Boeker [BB 67]. The lowest eigenstate for each value of I gives the ground state rotational band. It is in good agreement with projected HF wave functions. The excited states are vibrations. One finds anharmonicities in the spectra and for the moments and couplings between the different modes.

Faessler et al. [KRG 76] have investigated the monopole modes in the closed shell nuclei ^{16}O, ^{40}Ca, ^{90}Zr, and ^{208}Pb with constrained HF wave functions $\Phi(r_p, r_n)$ as generating states. The coordinates r_p, r_n are the rms radii for protons and neutrons, and the constraining operator (see Chap. 7) is given by

$$r^2\left(1+\exp\left(\frac{r-R}{a}\right)\right)^{-1}.$$

The fermi distribution makes sure that the constraining operator is bounded and the corresponding HF Hamiltonian has discrete eigenstates. The coupling to the continuum is neglected. The interaction is a modified δ-interaction of the Skyrme type, where the overlap integral $\mathcal{K}$ can be calculated in a simple way. With the constrained HF calculation, the scaling prescription can be checked and one finds that it is well justified. Equation (10.5) is again solved by discretization. The ground state energy of the GCM agrees very well with the HF energy. The excited states are breathing modes. Table 10.1 gives some results.

The energy of the isoscalar breathing mode in ^{208}Pb is 13.32 MeV and corresponds quite well to the liquid drop model with an incompressibility (Eq. 5.95) $K \simeq 200$ MeV and with the RPA calculation with the Migdal force [RS 74a].

The Skyrme interaction (see Sec. 4.4.3) in a basis of oscillator wave functions with variable oscillator parameters provides rather simple analytic expressions for $\mathcal{K}$ and $\mathcal{N}$. They were used by Giraud and Grammaticos [GG 75] in connection

Table 10.1. Results of GCM calculations for breathing modes: Excitation energy E, reduced transition probabilities, fraction S of the energy weighted sum rule (8.154), proton and neutron transition amplitudes A_p and A_n, isospin expectation value $\langle T^2 \rangle$, and proton and neutron radii r_p and r_n. (From [KRG 76].)

Nucleus	E (MeV)	B(E0) (fm^4)	S	A_p (fm^2)	A_n (fm^2)	$\langle T^2 \rangle$	r_p (fm)	r_n (fm)
^{16}O	0					0.00	2.70	2.67
	23.96	106	0.56	10.3	9.7	0.00	2.91	2.86
^{40}Ca	0					0.02	3.41	3.36
	20.73	502	0.55	22.4	20.4	0.14	3.50	3.44
^{90}Zr	0					30.12	4.22	4.30
	17.46	1530	0.45	39.1	49.8	30.12	4.25	4.33
^{208}Pb	0					506.38	5.46	5.66
	13.32	5749	0.36	75.8	127.4	506.41	5.47	5.67
	21.29	359	0.04	−19.0	130.6	506.59	5.45	5.71
	38.18	239	0.04	−15.4	26.3	506.68	5.44	5.79

with the moment expansion (see Sec. 10.7) for the description of monopole modes.

Flocard and Vautherin [FV 75, 76] solved the HW Eq. (10.5) for monopole, quadrupole, and dipole excitations with Slater determinants of oscillator wave functions and a Skyrme force by discretization (up to 143 mesh points). The ground state again agrees well with the HF state. Figure 10.2 shows weight functions g_i [Eq. (10.12)] for the ground state (g_0) and the first two excited breathing modes (g_1, g_2) in ^{16}O and the diagonal part of the energy overlap ($\mathcal{K}(r, r)$ in Eq. (10.135)), which corresponds more or less to the potential $V(r)$ of the collective coordinate. We see that the functions g behave qualitatively like oscillator functions in this potential.

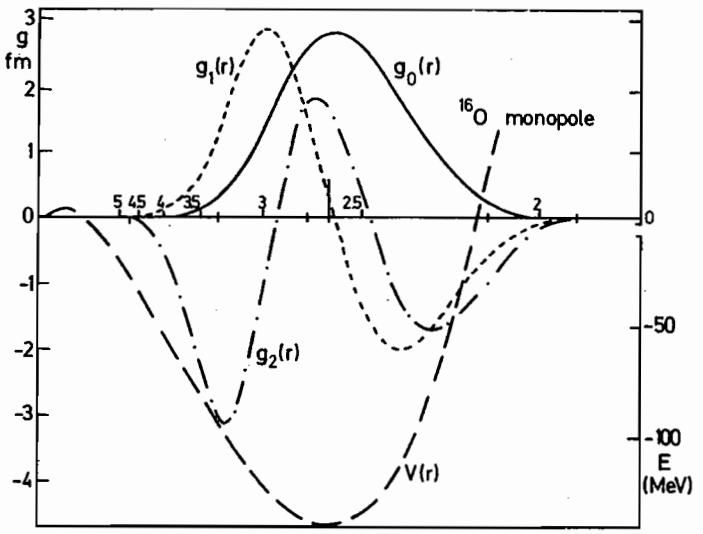


Figure 10.2. Weight functions g_i for the first three GCM states in ^{16}O in the case of monopole vibrations and the diagonal part of the energy overlap $V(r)$. (From [FV 76].)

For increasing mass number, the width of the function g_0 decreases. For heavy nuclei, it is sharply peaked at a certain value of the generator coordinate. This means that the GCM wave function is given, to a good approximation, by a Slater determinant with this parameter.

The GCM has also been applied to low-lying *shape vibrations* in light nuclei. As long as we restrict them to a configuration space which includes only an $s-d$ shell [GMA 75, MMR 75], they can be compared to an exact diagonalization of the Hamiltonian in this subspace [PW 72, Wh 72, SZ 72], and one finds good agreement for many levels. In some cases, however, the agreement is rather poor. This means that the corresponding state does not lie in the space spanned by the generating functions. A pure quadrupole constraint [KK 74] is then less effective than a constraining operator determined by the system itself, as discussed in Section 10.8 [PL 77].

10.9.2 Pairing Vibrations

In Sec. 8.8 we saw that in the description of pairing vibrations by the RPA based on a normal fluid ground state, we find a phase transition for a critical coupling constant G_c. In this region, neither the RPA nor the BCS model works well. The GCM provides a method to overcome this difficulty.

Several authors [JMR 72, ISY 73] proposed the use of BCS wave functions [see Eq. (6.31)] projected onto good particle number (see Sec. 11.4) as generating states

$$|\Phi^N\rangle = P^N \prod_{\alpha, m} (u_\alpha + v_\alpha a^+_{\alpha m} a^+_{\alpha -m})|-\rangle, \tag{10.157}$$

where P^N is the projection operator onto particle number N, and $\alpha=(nlj)$ are the quantum numbers of the different j-shells. For independent parameters $p_\alpha = u_\alpha/v_\alpha$, these states span the seniority-zero subspace of the full Hilbert space (see Sec. 6.2 and [JMR 69]). For occupation probabilities u_α, v_α of the BCS-form (6.59) that depend only on one parameter, Δ, this is still true to a good approximation. Since we expect the pairing vibrations with $J^\pi=0^+$ to lie in this subspace (see Sec. 8.8), the ansatz (10.157) is particularly useful for describing them within the GCM. The second parameter, the chemical potential λ, can be determined by the BCS-number equation (6.53) for each point in the Δ mesh, or can be used as a second generator coordinate (Sorensen et al. [SS 72b]).

This method works equally well for superconducting nuclei, non-superconducting nuclei, and in the intermediate region. It has been applied with the pure pairing force [SS 72b, FGP 73] to simple models and to pairing vibrational spectra in the Pb, the Ni, and Sn regions.

Figure 10.3 shows the components of the GCM wave function for the ground state in ^{208}Pb and for the so-called pairing vibrational state at 4.87 MeV, which is to a large extent a $2p-2h$ state or a superposition of the $2p$ pairing vibration in ^{210}Pb and of the $2h$-pairing vibration in ^{206}Pb. The $0p-0h$, $2p-2h$, and $4p-4h$ components are given as a function of the pairing strength G. The realistic value of G, which gives the right mass difference $M(^{206}\text{Pb})+M(^{210}\text{Pb})-2M(^{208}\text{Pb})$, is $G=0.1166$ (MeV). At this value, already 10% of the ground state is $2p-2h$ and the pairing vibration has roughly 12% $4p-4h$ admixtures.

More realistic forces have been used by Faessler et al. [AGM 74] in the $p-f$ shell. Here the BCS functions in Eq. (10.157) were replaced by constrained HFB

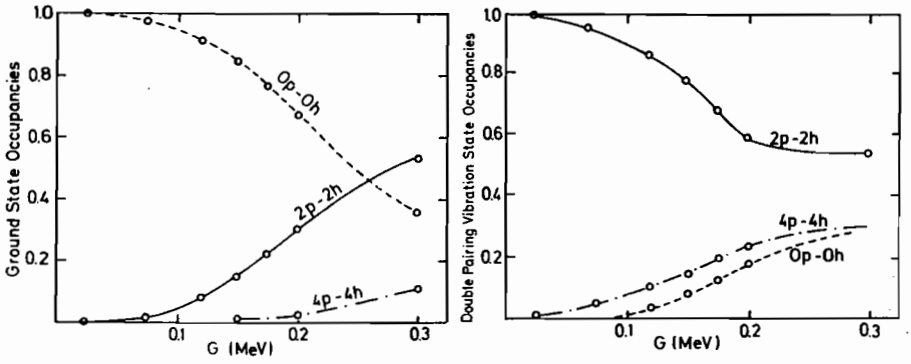


Figure 10.3. Percentage of ^{208}Pb ground state and pairing vibrational state at 4.87 MeV, which is $0p-0h$, $2p-2h$, and $4p-4h$ as a function of G. (From [SS 72b].)

functions (see Sec. 7.6). The constraining operator is a pairing potential of variable strength and the generator coordinate is the resulting energy contribution $-\frac{1}{4}\mathrm{Tr}(\kappa^*\bar{v}\kappa)$. The generating functions were determined from these HFB functions by projection onto good particle number and angular momentum. The results are in good agreement with experimental energy levels and transfer cross sections.

This method provides a tool for investigating the dependence of the pairing correlations (measured by the energy gap Δ) on the effective pairing strength G_{eff} in the ground and pairing vibrational states (Fig. 10.4). In cases where the ground state is strongly paired (large G_{eff}), the pairing vibration has weak pairing correlations. This is the case, for instance, in the rare earth nuclei. The opposite is true for nuclei with weak pairing in the ground state, as in Ti (see also [Kl 70]). From Section 7.7 we know that the Coriolis antipairing effect (CAP) reduces the effective pairing strength with increasing angular momentum. This produces decreasing pairing correlations in the ground state of the rare earth nuclei, and increasing pairing correlations in the pairing vibrational mode. The same is no longer true, however, for lighter nuclei such as Ti.

Finally, we can describe various couplings between different modes by the GCM, if we introduce more generator coordinates, such as deformation parameters

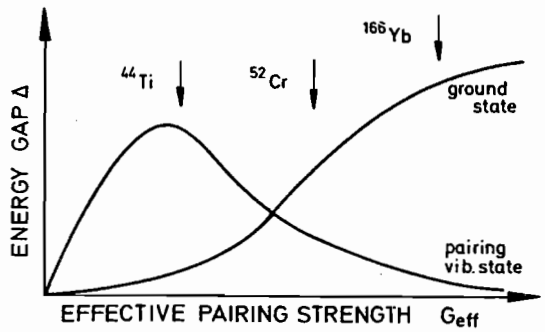


Figure 10.4. Schematic picture of the dependence of the energy gap Δ on the effective pairing strength G_{eff}. The position is indicated by arrows for a few nuclei. (From [GSF 75].)

β, pairing parameters Δ, and so on [MAG 75]. We can even include *single-particle degrees of freedom* by making the generalized ansatz [DMG 76, MGA 77]:

$$|\Psi\rangle = \int da\, f(a)|\Phi(a)\rangle + \sum_i \int da\, f^i(a)|\Phi_i(a)\rangle, \tag{10.158}$$

where $|\Phi(a)\rangle$ are, for instance, the ground states of a constrained HF calculation dependent on some deformation parameter a and $|\Phi_i(a)\rangle$ are the corresponding $1p-1h$ states, and so on.